A NON-B-SEMI-AMPLE MODULI PART OF AN LC-TRIVIAL FIBRATION WITH NON-EFFECTIVE BOUNDARY

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Abstract. We show that there exists an lc-trivial fibration of relative dimension 1 whose boundary is not effective over the generic point of the base and whose moduli part is not b-semi-ample, so the effectivity hypothesis in the recent b-semi-ampleness theorem for lc-trivial fibrations cannot be removed. We also show that, in the formulation of Han, Jiang, and Luo, which allows non-effective boundaries, Shokurov's conjecture on singularities on the base of a Fano type fibration fails in relative dimension 2.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let $\pi : (X, B) \to Z$ be an lc-trivial fibration (see Definition 2.4), with discriminant $\boldsymbol{b}$-divisor $\mathbf{B}$ and moduli $\boldsymbol{b}$-divisor $\mathbf{M}$ (see Definition 2.5). Then

$$K_X + B \sim_{\mathbb{R}} \pi^*(K_Z + B_Z + M_Z).$$

In the following results of [Kaw97, Kaw98, Amb04, Amb05, PS09, FG14], B is a $\mathbb{Q}$-divisor. Kawamata proved that $\mathbf{M}$ is $\boldsymbol{b}$-semi-ample when $\dim X - \dim Z = 1$ and $B \geq 0$ over the generic point of Z [Kaw97] (see [BFMT25, Section 1.2.1]), and that $\mathbf{M}_{Z'}$ is nef on some higher model Z' of Z [Kaw98]. Ambro introduced lc-trivial fibrations, in which $B \geq 0$ is not assumed, under the condition that (X, B) is sub-klt over the generic point of Z [Amb04, Definition 2.1(1)]. When (X, B) is sub-klt over the generic point of Z, $\mathbf{M}$ descends to some higher model Z' of Z and $\mathbf{M}_{Z'}$ is nef [Amb04, Theorem 0.2], and M_Z is semi-ample when $\dim Z = 1$ [Amb04, Theorem 0.1]. Neither result assumes $B \geq 0$. If moreover $B \geq 0$ over the generic point of Z, then there exist a higher model Z' of Z to which $\mathbf{M}$ descends, a contraction $h : Z' \to Z''$, and a nef and big $\mathbb{Q}$-divisor N on Z'' such that

$$\mathbf{M}_{Z'} \sim_{\mathbb{Q}} h^*N$$

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[Amb05, Theorem 3.3 and Definition 3.2]. Under the conditions of [PS09, 7.1 and Assumption 7.11], which include $B \geq 0$ over the generic point of Z, Prokhorov and Shokurov conjectured that $\mathbf{M}$ is $\boldsymbol{b}$-semi-ample [PS09, Conjecture 7.13], and proved the conjecture when $\dim X = \dim Z + 1$ using a construction and a result of [Kaw97] [PS09, Theorem 8.1]. Fujino and Gongyo extended the descent and the nefness of $\mathbf{M}$ to the case when (X, B) is only sub-lc over the generic point of Z [FG14, Theorem 3.6] (see [FG14, Definition 3.2(1)]). Recently, [BFMT25] proved the following theorem.

Theorem 1.1 ([BFMT25, Theorem 1.5 and Remark 6.30])**.** *Let $\pi : (X, B) \to Z$ be an lc-trivial fibration such that B is a $\mathbb{Q}$-divisor and $B \geq 0$ over the generic point of Z. Then the moduli $\boldsymbol{b}$-divisor of π is $\boldsymbol{b}$-semi-ample.*

In [BFMT25], lc-trivial fibrations are defined for sub-pairs with coefficients in $\mathbb{Q}$ [BFMT25, Definition 6.18], and [BFMT25, Remark 6.30] shows that the $\boldsymbol{b}$-semi-ampleness of the moduli part of [BFMT25] is equivalent to the $\boldsymbol{b}$-semi-ampleness of $\mathbf{M}$. We ask whether the condition $B \geq 0$ over the generic point of Z can be removed from Theorem 1.1.

Shokurov conjectured that, when the fibers of π are of Fano type, the singularities of (X, B) bound the singularities of the base. In [AB14, Conjecture 1.2] and [Bir16, Conjecture 1.2], as read in [HJL22, Remark 1.6(1)], the conjecture assumes that $B \geq 0$ and (X, B) is ϵ-lc (see Definition 8.1), where X is of Fano type over Z in [AB14] and the general fibers of π are of Fano type in [Bir16]. It predicts that there exists an $\mathbb{R}$-divisor $D \geq 0$ on Z with $D \sim_{\mathbb{R}} M_Z$ such that $(Z, B_Z + D)$ is δ-lc, where $\delta > 0$ depends only on $\dim X$ and ϵ. Han, Jiang, and Luo formulated the conjecture over a point $z \in Z$ in terms of the minimal log discrepancy $\mathrm{mld}(X/Z \ni z, B)$ (see Definition 8.1).

Conjecture 1.2 ([HJL22, Conjecture 1.5])**.** *Let d be a positive integer and ϵ a positive real number. Then there exists a positive real number δ depending only on d and ϵ satisfying the following. Let $\pi : (X, B) \to Z$ be an lc-trivial fibration and $z \in Z$ a (not necessarily closed) point of codimension ≥ 1 such that*

(1) $\dim X - \dim Z = d$,
(2) $\mathrm{mld}(X/Z \ni z, B) \geq \epsilon$, and
(3) the generic fiber of π is of Fano type.

Then there exists an $\mathbb{R}$-divisor $D \geq 0$ on Z with $D \sim_{\mathbb{R}} M_Z$ such that $(Z \ni z, B_Z + D)$ is δ-lc.

Conjecture 1.2 assumes condition (2) in place of the condition in [AB14, Bir16] that $B \geq 0$ and (X, B) is ϵ-lc, so B may be non-effective and (X, B) may have non-klt centers over $Z \backslash \{z\}$ [HJL22, Remark 1.6(1)]. Conjecture 1.2 was known in the following cases:

(1) When $d = 1$ [CH21] (see [HJL22, Remark 1.6(3)]).
(2) When $d = 1$ and $\epsilon \geq 1$, with $\delta = \epsilon - \frac{1}{2}$ [HJL22].

When $B \geq 0$, (X, B) is ϵ-lc, and the couples $(F, \operatorname{Supp} B|_F)$, for the general fibers F of π, belong to a bounded family, the conclusion of Conjecture 1.2 holds with δ depending only on $\dim X$, ϵ, and the bounded family [Bir16]. When $d = 1$, the generic fiber of π is a conic, and the rank condition in Definition 2.4 forces $B \geq 0$ over the generic point of Z (see [HJL22, Remark 2.13]), so in Conjecture 1.2, B can be non-effective over the generic point of Z only when $d \geq 2$.

For $B \geq 0$, Birkar proved a generalized version: if B is a $\mathbb{Q}$-divisor, $B \geq 0$, (X, B) is ϵ-lc, and X is of Fano type over Z, then the generalized pair $(Z, B_Z, \mathbf{M})$ is generalized δ-lc, where $\delta > 0$ depends only on $\dim X - \dim Z$ and ϵ [Bir23, Theorem 1.1]. Birkar and Chen proved a local version for $\mathbb{R}$-divisors: if $T \subset Z$ is a closed subvariety, $B \geq 0$, X is of Fano type over Z, and (X, B) is ϵ-lc along $\pi^{-1}T$, then $(Z, B_Z, \mathbf{M})$ is generalized δ-lc along T, where $\delta > 0$ depends only on $\dim X - \dim Z$ and ϵ [BC25, Theorem 1.3] (see [BC25, Definition 1.1]). When $d = 1$ and $0 < \epsilon \leq 1$, [BC25, Theorem 1.3] was proved earlier with an optimal δ [Che25, Theorem 1.4] (see [BC25, Remark 1.4(1)]).

We ask whether Conjecture 1.2, in which B may be non-effective, holds when $d = 2$.

Main results.

Theorem 1.3. *There exists an lc-trivial fibration $\pi : (X, B) \to Z$ with B a $\mathbb{Q}$-divisor satisfying the following.*

(1) X and Z are smooth projective varieties with $\dim Z = 5$ and $\dim X - \dim Z = 1$.
(2) The generic fiber X_η of π is a smooth curve of genus 1, and $B|_{X_\eta} = \frac{1}{5}D_\eta - P_\eta$ for a $\mathbb{C}(Z)$-rational point P_η of X_η and a reduced divisor D_η on X_η with $D_\eta \sim 5P_\eta$ and $P_\eta \notin \operatorname{Supp} D_\eta$.
(3) The moduli $\boldsymbol{b}$-divisor $\mathbf{M}$ of π descends to Z, and M_Z is nef and not semi-ample.

In particular, $\mathbf{M}$ is not $\boldsymbol{b}$-semi-ample.

By Theorem 1.3(2), B is not ≥ 0 over the generic point of Z, while (X, B) is sub-klt over the generic point of Z. Thus the condition that $B \geq 0$ over the generic point of Z cannot be removed from Theorem 1.1, even when B is a $\mathbb{Q}$-divisor, X and Z are smooth projective, and $\dim X - \dim Z = 1$.

For lc-trivial fibrations of relative dimension 2 with generic fiber of Fano type, we obtain the following.

Corollary 1.4. *There exist an lc-trivial fibration $\pi : (X, B) \to Z$ with B a $\mathbb{Q}$-divisor and a closed point $z \in Z$ satisfying the following.*

(1) $\dim X - \dim Z = 2$.
(2) $\operatorname{mld}(X/Z \ni z, B) \geq 1$.
(3) The generic fiber of π is of Fano type.
(4) For any open neighborhood $V \subset Z$ of z and any $\mathbb{R}$-divisor $D \geq 0$ on V with $D \sim_{\mathbb{R}} M_Z|_V$, we have $\operatorname{mld}(V \ni z, B_Z|_V + D) = -\infty$.

In particular, Conjecture 1.2 does not hold for $d = 2$ and $\epsilon = 1$.

Remark 1.5. In Corollary 1.4, we can take $\pi : (X, B) \to Z$ with generic fiber the weighted projective plane $S := \mathbb{P}(1, 2, 3)$ over the function field of Z and with the restriction of B to the generic fiber equal to $E_S + \frac{1}{5}D_S - H$, where H is the unique member of $|\mathcal{O}_S(1)|$, $E_S \in |6H|$ is a smooth curve, and $D_S \in |5H|$ is a curve not containing H (see Section 8). For such π, B is not ≥ 0 over the generic point of Z, and (X, B) is not sub-klt over the generic point of Z. So π does not satisfy the hypotheses of [AB14, Conjecture 1.2] and [Bir16, Conjecture 1.2], as read in [HJL22, Remark 1.6(1)], or of [BC25, Theorem 1.3]. Moreover, we construct such π from an lc-trivial fibration whose generic fiber is S with boundary $E_S + \frac{1}{5}D_S - H$ and whose moduli $\boldsymbol{b}$-divisor is not $\boldsymbol{b}$-semi-ample (see Construction-Proposition 7.1 and Proposition 5.1). By Theorem 1.1, the moduli $\boldsymbol{b}$-divisor of an lc-trivial fibration is $\boldsymbol{b}$-semi-ample when the boundary is a $\mathbb{Q}$-divisor that is ≥ 0 over the generic point of the base.

Sketch of the proof. Let U be the smooth variety of dimension 5 parametrizing the curves E_u of genus 1 with points P_u and reduced divisors $D_u \sim 5P_u$ of Construction-Proposition 4.1. Over a dense open subset $U' \subset Z$, the fibration $\pi : (X, B) \to Z$ of Theorem 1.3 is the base change of the family of $(E_u, \frac{1}{5}D_u - P_u)$ by a finite étale morphism g from U' to an open subset of U (see Section 6). For $(E_u, \frac{1}{5}D_u - P_u)$, the rank condition of Definition 2.4(2) is $h^0(E_u, \mathcal{O}_{E_u}(P_u)) = 1$, since $\lceil \mathbf{A}^*(E_u, \frac{1}{5}D_u - P_u) \rceil = P_u$ (see Definition 2.3 and Section 6), whereas on a conic it forces the boundary to be ≥ 0 (see [HJL22, Remark 2.13]). The canonical bundle formula of [BFMT25, Construction-Definition 6.23 and Theorem 6.28], which does not assume $B \geq 0$, computes $\mathbf{M}$ from the cyclic covers $c_u : C_u \to E_u$ of degree 5 branched along D_u. For the character χ of μ_5 of Construction-Proposition 4.1, the χ-isotypic part of $H^1(C_u, \mathbb{C})$ has Hodge numbers $(1, 4)$, and its $(1, 0)$-parts $H^0(E_u, \omega_{E_u}(P_u))$ form a line bundle $\mathcal{A}$ on U. For the Schmid extension $\mathcal{A}_Z$ [BFMT25, Construction-Definition 6.23] of $g^*\mathcal{A}$, $\mathbf{M}$ descends to Z and

$$\mathcal{O}_Z(mM_Z) \cong \mathcal{A}_Z^{\otimes m}$$

for some $m \in \mathbb{N}^+$ (see Proposition 3.3). Thus Theorem 1.3(3) holds since $\mathcal{A}_Z$ is nef and not semi-ample (see Proposition 5.1). For Corollary 1.4, $\mathcal{A}$ also determines the moduli part of a family of sub-pairs on $\mathbb{P}(1,2,3)$ (see Construction-Proposition 7.1).

Let $\mathbb{V}$ be the variation of Hodge structure on U with fibers

$$\mathbb{V}_u = \operatorname{Ker}\big(c_{u*} : H^1(C_u, \mathbb{Q}) \to H^1(E_u, \mathbb{Q})\big),$$

so that $\mathcal{A} \subset \mathbb{V}_\chi \otimes \mathcal{O}_U$ for the χ-isotypic part $\mathbb{V}_\chi$ of $\mathbb{V} \otimes \mathbb{C}$ (see Construction-Proposition 4.1). After any dominant generically finite base change, the period map of $\mathbb{V}$ is immersive at a general point, and the connected algebraic monodromy group of $\mathbb{V}$ acts irreducibly on $\mathbb{V}_u$ (see Proposition 4.2). By Griffiths' theorem [Voi02, Theorem 10.21] and μ_5-equivariant Serre duality on C_u, the immersivity reduces to the surjectivity of

$$H^0(E_u, \mathcal{O}_{E_u}(2P_u)) \otimes H^0(E_u, \mathcal{O}_{E_u}(3P_u)) \to H^0(E_u, \mathcal{O}_{E_u}(5P_u)),$$

which holds by the base-point-free pencil trick [ACGH85, p. 126], since $H^1(E_u, \mathcal{O}_{E_u}(P_u)) = 0$. For the irreducibility, [Rai25, Theorem 9.2], on μ_N-covers of curves of any genus, implies that the covers of E_u in which one point of D_u moves have large monodromy [Rai25, Definition 3.1], so the connected algebraic monodromy group contains $\operatorname{Res}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}} SU$ of the anti-Hermitian form on $\mathbb{V}_u$ over $\mathbb{Q}(\zeta_5)$ in that definition, and $\operatorname{Res}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}} SU$ acts irreducibly on $\mathbb{V}_u$. The immersivity is seen on $\mathbb{V}_{\chi^2}$, a conjugate of $\mathbb{V}_\chi$ under $\operatorname{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q})$ other than $\overline{\mathbb{V}_\chi}$, whose Hodge numbers are $(2,3)$: by the surjectivity above, the map

$$T_{U,u} \to \operatorname{Hom}(\mathbb{V}^{1,0}_{\chi^2,u}, \mathbb{V}^{0,1}_{\chi^2,u}),$$

from a space of dimension 5 to one of dimension 6, given by the Gauss–Manin connection is injective.

Shifting the Hodge types of $\mathbb{V}_\chi$ by $(2,-2)$ and those of $\mathbb{V}_{\chi^{-1}} = \overline{\mathbb{V}_\chi}$ by $(-2,2)$, and keeping those of $\mathbb{V}_{\chi^2}$ and $\mathbb{V}_{\chi^3}$, yields a polarized real variation of Hodge structure $\mathbb{V}'$ with $F^3\mathbb{V}' = \mathcal{A}$ and $F^2\mathbb{V}' = \mathbb{V}_\chi \otimes \mathcal{O}_U$ (compare [BFMT25, Remark 6.26]), since the shifts of $\mathbb{V}_\chi$ and $\overline{\mathbb{V}_\chi}$ are complex conjugate and keep $(\sqrt{-1})^{p-q}$ in the Hodge–Riemann bilinear relations. Thus, by [BBT23, Lemma 6.17], $\mathcal{A}_Z$ is nef, and it is big if and only if the map

$$\psi : T_{U'} \to \operatorname{Hom}\big(g^*\mathcal{A}, (g^*\mathbb{V}_\chi \otimes \mathcal{O}_{U'})/g^*\mathcal{A}\big)$$

given by the Gauss–Manin connection is injective. Since $\operatorname{rank}\psi \le \operatorname{rank}\mathbb{V}_\chi - 1 = 4 < 5 = \dim U'$, ψ is not injective, and $\mathcal{A}_Z$ is not big.

Suppose that $\mathcal{A}_Z$ is semi-ample, and let Z_y be a general fiber of the morphism defined by $|\mathcal{A}_Z^{\otimes n}|$ for some $n \in \mathbb{N}^+$. Then $\dim Z_y \ge 1$ since $\mathcal{A}_Z$ is not big, and ψ vanishes on the tangent spaces of $Z_y \cap U'$ (see (5.1)), so the flat sections of $g^*\mathcal{A}|_{Z_y \cap U'}$ form a rank 1 local subsystem of $g^*\mathbb{V} \otimes \mathbb{C}$ with finite monodromy [Del71, Corollaire (4.2.8)(iii)(b)]. The monodromy group of $g^*\mathbb{V}|_{Z_y \cap U'}$ is normal in that of $g^*\mathbb{V}$, so it is finite (see Proposition 4.2(2)), and the Hodge structure of $g^*\mathbb{V}$ is locally constant along $Z_y \cap U'$, which contradicts Proposition 4.2(1).

The family of Construction-Proposition 7.1 has base W (written Z there) and generic fiber S with boundary $E_S + \frac{1}{5}D_S - H$ as in Remark 1.5. Over a dense open subset of U, a Weierstrass equation embeds E_u into S as $E_S \in |6H|$ with $H|_{E_S} = P_u$, and $D_u = D_S|_{E_S}$ for a unique $D_S \in |5H|$, since the restriction $H^0(S, \mathcal{O}_S(5H)) \to H^0(E_u, \mathcal{O}_{E_u}(5P_u))$ is an isomorphism. Then $K_S + E_S + \frac{1}{5}D_S - H \sim_{\mathbb{Q}} 0$, and by adjunction,

$$\Big(K_S + E_S + \frac{1}{5}D_S - H\Big)\Big|_{E_S} = K_{E_u} + \frac{1}{5}D_u - P_u.$$

The rank condition for $E_S + \frac{1}{5}D_S - H$ is $h^0(S, \mathcal{O}_S(H)) = 1$, which holds since H is the unique member of $|\mathcal{O}_S(1)|$. The cyclic cover of S of degree 5 branched along D_S contains C_u over E_S, and it is a rational surface since the equation of D_S has degree 1 in the coordinate of weight 3 (see (7.1)). Thus, in the χ-isotypic part of H^2 of the complement of the preimage of C_u in a resolution of the cover (see Set-up 3.1), the weight 2 part has $F^2 = 0$, and the

residue along C_u is an isomorphism from the weight 3 part onto $H^1(C_u,\mathbb{C})_\chi(-1)$. With the equality in [BFMT25, Construction-Definition 6.23] that passes to the transcendental part, which reduces the Schmid extension of F^2 to the weight 3 part, the canonical bundle formula implies that $\mathbf{M}$ descends to W and $\mathcal{O}_W(mM_W)$ is, for some $m\in\mathbb{N}^+$, the m-th tensor power of the Schmid extension of $g^*\mathcal{A}$ for the morphism $g:U'\to U$ of Construction-Proposition 7.1 (see Proposition 3.3 and Construction-Proposition 7.1(2)). So M_W is nef, not big, and not semi-ample (see Proposition 5.1).

In Section 8, we fix $p\in W$ with $p\in\operatorname{Supp}D$ for any $\mathbb{R}$-divisor $D\geq 0$ with $D\sim_{\mathbb{R}} M_W$, which exists since M_W is not semi-ample, and replace the discriminant of the family by one for which p is a 0-dimensional lc center and $\mathbf{M}$ does not change. Let μ contract the zero section $E\cong W$ of the total space of $\mathcal{O}_W(-A)$, for a very ample divisor A, to the vertex o of the cone Z_0 over (W,A) [Kol13, 3.8]. The fibration $\pi:(X,B)\to Z$ of Corollary 1.4 has base $Z=Z_0\times\mathbb{A}^1$, and $z=(o,0)$. For an open neighborhood V of z and an $\mathbb{R}$-divisor $D\geq 0$ on V with $D\sim_{\mathbb{R}} M_Z|_V$, the crepant pullback of $(V,B_Z|_V+D)$ by $\mu\times\mathrm{id}_{\mathbb{A}^1}$ has coefficient $1+\alpha$ along $E\times\mathbb{A}^1$ for some $\alpha\in\mathbb{R}$, and the strict transform $\widetilde{D}$ of D satisfies

$$\widetilde{D}|_{E\times\{s\}}\sim_{\mathbb{R}} M_W+\alpha A\quad\text{for general } s$$

(see (8.7)). If $\alpha<0$, then M_W is big, a contradiction. If $\alpha\geq 0$, then $\operatorname{mld}(V\ni z,B_Z|_V+D)=-\infty$, since $1+\alpha>1$ when $\alpha>0$, and $\{p\}\times\mathbb{A}^1\subset\operatorname{Supp}\widetilde{D}$ is an lc center of the boundary without $\widetilde{D}$ when $\alpha=0$.

In [BFMT25, Examples 4.6 and 4.7], shifting the Hodge types of two complex conjugate summands as for $\mathbb{V}'$ yields polarized variations whose Hodge bundles are not semi-ample, and [BFMT25, Remark 4.8] says that neither variation is manifestly algebraic. Here the local system $\mathbb{V}\otimes\mathbb{R}$ of $\mathbb{V}'$ is a direct summand of the first cohomology local system of the family of curves C_u.

Structure of the paper. Section 2 recalls lc-trivial fibrations, their discriminants, and their moduli parts. Section 3 shows that, under conditions that hold after shrinking the base, a finite étale base change, and a compactification, the moduli part is $\mathbb{Q}$-linearly equivalent to the Schmid extension of a Hodge line bundle of a family of cyclic covers. Section 4 constructs a family of cyclic covers of degree 5 of curves of genus 1, and studies the period map and the monodromy group of the family. Section 5 proves that the Schmid extension of a Hodge line bundle of the family of Section 4 is nef, not big, and not semi-ample, and Section 6 proves Theorem 1.3. Section 7 constructs a family of sub-pairs on the weighted projective plane $\mathbb{P}(1,2,3)$ whose moduli part is $\mathbb{Q}$-linearly equivalent to the Schmid extension of the Hodge line bundle of Section 4, and Section 8 proves Corollary 1.4.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and singularities.

Definition 2.1. A *pair* $(X/Z\ni z,B)$ consists of a contraction $\pi:X\to Z$ between normal quasi-projective varieties, a point $z\in Z$, and an $\mathbb{R}$-divisor $B\geq 0$ on X such that K_X+B is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X\ni z,B)$ is a pair. If $(X/Z\ni z,B)$ is a pair for any $z\in Z$, then we say that $(X/Z,B)$ is a pair and also say that (X,B) is a pair. If we do not require that $B\geq 0$, then we say that

$(X/Z \ni z, B)$ is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B).

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Let $\pi : X \to Z$ be a morphism, (X, B) a sub-pair, and $T \subset Z$ a closed irreducible subset. We say that (X, B) is *sub-lc over* an open subset $V \subset Z$ if $(\pi^{-1}(V), B|_{\pi^{-1}(V)})$ is sub-lc, and *sub-lc over the generic point of* T if (X, B) is sub-lc over some open subset $V \subset Z$ with $V \cap T \neq \emptyset$. We say that $B \geq 0$ *over the generic point of* T if $B|_{\pi^{-1}(V)} \geq 0$ for some open subset $V \subset Z$ with $V \cap T \neq \emptyset$.

2.2. **Lc-trivial fibrations.**

Definition 2.2. Let Z be a normal variety. A ***b**-divisor* $\mathbf{D}$ over Z assigns to each projective birational morphism $Z' \to Z$ from a normal variety an $\mathbb{R}$-divisor $\mathbf{D}_{Z'}$ on Z', the *trace* of $\mathbf{D}$ on Z', such that $g_*\mathbf{D}_{Z''} = \mathbf{D}_{Z'}$ for any projective birational morphism $g : Z'' \to Z'$ over Z. We say that $\mathbf{D}$ *descends* to Z' if $\mathbf{D}_{Z'}$ is $\mathbb{R}$-Cartier and $\mathbf{D}_{Z''} = g^*\mathbf{D}_{Z'}$ for any such g. We say that $\mathbf{D}$ is ***b**-semi-ample* if $\mathbf{D}$ descends to some Z' with $\mathbf{D}_{Z'}$ a semi-ample $\mathbb{Q}$-divisor.

For a ***b***-divisor $\mathbf{D}$ over Z, $\lceil \mathbf{D} \rceil$ is the ***b***-divisor with traces $\lceil \mathbf{D}_{Z'} \rceil$, and $\mathcal{O}_Z(\mathbf{D})$ is the sheaf on Z whose sections over an open subset $V \subset Z$ are 0 and the rational functions $\varphi \neq 0$ on Z with $\operatorname{div}(\varphi) + \mathbf{D}_{Z'} \geq 0$ over the preimage of V for any projective birational morphism $Z' \to Z$ from a normal variety.

Definition 2.3. For a sub-pair (X, B), the *discrepancy* ***b**-divisor* $\mathbf{A}(X, B)$ over X has trace

$$\mathbf{A}(X, B)_{X'} := K_{X'} - h^*(K_X + B)$$

on X' for any projective birational morphism $h : X' \to X$ from a normal variety, where $K_{X'}$ is the canonical divisor with $h_*K_{X'} = K_X$, so $\operatorname{mult}_E \mathbf{A}(X, B)_{X'} = a(E, X, B) - 1$ for any prime divisor E on X', and

$$\mathbf{A}^*(X, B)_{X'} := \sum_{\substack{E \subset X' \text{ prime divisor} \\ a(E,X,B)>0}} (a(E, X, B) - 1)E.$$

Definition 2.4 (cf. [HJL22, Definition 2.12] and [BFMT25, Definition 6.18])**.** An *lc-trivial fibration* $\pi : (X, B) \to Z$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties and a sub-pair (X, B) satisfying the following.

(1) (X, B) is sub-lc over the generic point of Z.
(2) $\operatorname{rank} \pi_*\mathcal{O}_X(\lceil \mathbf{A}^*(X, B) \rceil) = 1$.
(3) There exists an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor L on Z with $K_X + B \sim_{\mathbb{R}} \pi^*L$.

Definition 2.5 (cf. [HJL22, (2.1)–(2.2)])**.** Let $\pi : (X, B) \to Z$ be an lc-trivial fibration and L as in Definition 2.4. If B is a $\mathbb{Q}$-divisor, then L is taken to be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor with $K_X + B \sim_{\mathbb{Q}} \pi^*L$, and L, hence M_Z below, is unique up to $\mathbb{Q}$-linear equivalence.

For a prime divisor P on Z, let t_P be the supremum of the real numbers t such that $(X, B+t\pi^*P)$ is sub-lc over the generic point of P, where π^*P is taken over an open subset of Z meeting P on which P is Cartier. The *discriminant* and the *moduli part* of π are

$$B_Z := \sum_P (1 - t_P)P \quad \text{and} \quad M_Z := L - K_Z - B_Z.$$

For a projective birational morphism $g : Z' \to Z$ from a normal variety, let X' be a resolution of the main component of $X \times_Z Z'$ with induced morphisms $h : X' \to X$ and $\pi' : X' \to Z'$,

and $K_{X'} + B' := h^*(K_X + B)$. Since $\mathbf{A}^*(X', B') = \mathbf{A}^*(X, B)$ and g is an isomorphism over a non-empty open subset of Z, $\pi' : (X', B') \to Z'$ is an lc-trivial fibration. The discriminant and the moduli part of $\pi' : (X', B') \to Z'$, with g^*L in place of L, are the traces on Z' of the *discriminant* $\boldsymbol{b}$*-divisor* $\mathbf{B}$ and the *moduli* $\boldsymbol{b}$*-divisor* $\mathbf{M}$ of π.

The $\boldsymbol{b}$-divisor $\mathbf{A}^*(X, B)$ of [BFMT25, Example 6.1] is $\mathbf{A}(X, B)$ with the coefficients of the prime divisors E with $a(E, X, B) = 0$ set to 0, which coincides with $\mathbf{A}^*(X, B)$ of Definition 2.3 over the preimage of any open subset $V \subset Z$ over which (X, B) is sub-lc, so under Definition 2.4(1), under which a non-empty such V exists, Definition 2.4(2) is [BFMT25, Definition 6.18(ii)]. For a $\mathbb{Q}$-divisor B, the convention on L in Definition 2.5 is the one of [BFMT25, Definition 6.18]. The number $1 - t_P$ is the supremum in [BFMT25, Definition 6.20] (the number $1 + a(E; Y, \Delta)$ in [BFMT25, Definition 6.20] equals $a(E, X, B)$ in our notation), that is, B_Z is the $\mathbb{Q}$-divisor B_X of [BFMT25, Definition 6.20]. The $\mathbb{Q}$-divisor M_Z differs from the Shokurov moduli divisor of [BFMT25, Definition 6.21] only by $\mathbb{Q}$-linear equivalence.

3. The moduli part as a Hodge bundle

Set-up 3.1. Let $\pi : (X, B) \to Z$, Σ, ϕ, and h be as follows.

(1) $\pi : (X, B) \to Z$ is an lc-trivial fibration of relative dimension d between normal projective varieties, Z is smooth, and B is a $\mathbb{Q}$-divisor.
(2) $\Sigma \subset Z$ is an snc divisor such that, for $Z^\circ := Z \setminus \Sigma$, $X^\circ := \pi^{-1}(Z^\circ)$, and $B^\circ := B|_{X^\circ}$, the variety X° is smooth, $\operatorname{Supp} B^\circ$ is an snc divisor, and $(X^\circ, \operatorname{Supp} B^\circ)$ is topologically locally trivial over Z°.
(3) ϕ is a rational function on X such that
$$r(K_{X^\circ} + B^\circ) + \operatorname{div}(\phi)|_{X^\circ} = 0$$
for a canonical divisor K_{X° of X°, where X_η is the generic fiber of π and
$$r := \min\{k \in \mathbb{N}^+ \mid k(K_{X_\eta} + B|_{X_\eta}) \sim 0\}.$$
(4) $f : X' \to X^\circ$ is the normalization of the cyclic cover of X° given by an r-th root $\phi^{1/r}$ of ϕ, with the induced action of μ_r, $(B^\circ)^{=1}$ is the sum of the components of B° with coefficient 1, and $h : X'' \to X'$ is a μ_r-equivariant resolution such that the induced morphism $\pi'' : X'' \to Z^\circ$ is smooth and the reduced inverse image E'' of $(B^\circ)^{=1}$ on X'' is an snc divisor each of whose strata is smooth over Z°.
(5) $R^d\pi''_*\mathbb{Z}_{X'' \setminus E''}$ has unipotent local monodromy around Σ.

Let
$$\mathcal{L} := \mathcal{O}_{X^\circ}(-K_{X^\circ} - \lfloor B^\circ \rfloor),$$
let χ be the character of μ_r with $\zeta \cdot \phi^{1/r} = \chi(\zeta)\phi^{1/r}$ for any $\zeta \in \mu_r$, let $\mathbb{H} := R^d\pi''_*\mathbb{C}_{X'' \setminus E''}$, and let $\mathbb{H}_\chi$ be the χ-isotypic part of $\mathbb{H}$. We denote by $(F^d\mathbb{H}_\chi)_Z$ the Schmid extension [BFMT25, Construction-Definition 6.23] of the deepest Hodge piece $F^d\mathbb{H}_\chi$ of $\mathbb{H}_\chi$.

Conditions (1)–(5) of Set-up 3.1 imply the conditions (†) and (††) of [BFMT25, Construction-Definition 6.23] for the cyclic cover given by an r-th root of ϕ and the resolution h, and $\mathbb{H}_\chi$ is the χ-isotypic part of the variation of mixed Hodge structures $V_Y \otimes \mathbb{C}$ there.

Lemma 3.2 (cf. [BFMT25, Construction-Definition 6.23] and [FG14, Remark 3.9])**.** *Notation as in Set-up 3.1, assuming only conditions (1)–(4) there, and let* $G := (B^\circ)^{=1} - \lfloor B^\circ \rfloor$*. Then the* χ*-isotypic summand of* $f_*\omega_{X'}(f^*(B^\circ)^{=1})$ *is*
$$\omega_{X^\circ}((B^\circ)^{=1}) \otimes \mathcal{L} \cong \mathcal{O}_{X^\circ}(G),$$
and $F^d\mathbb{H}_\chi$ *is a line bundle on* Z° *with*
$$F^d\mathbb{H}_\chi \cong \pi_*\mathcal{O}_{X^\circ}(G) \otimes \omega_{Z^\circ}^{-1}.$$

Proof. Write $B^\circ = \sum_i b_i B_i$, where the B_i are distinct prime divisors, and let $D' := f^*(B^\circ)^{=1}$. Since $(X^\circ, \operatorname{Supp} B^\circ)$ is topologically locally trivial over Z°, any B_i dominates Z°, so $b_i \le 1$ for any i by Definition 2.4(1).

By the minimality of r, $\deg f = r$, and by [EV92, Corollary 3.11],

$$f_*\mathcal{O}_{X'} = \bigoplus_{i=0}^{r-1} \mathcal{L}^{-i}(\lfloor i\{B^\circ\}\rfloor) \cdot \phi^{-i/r},$$

where $\phi^{-i/r} := (\phi^{1/r})^{-i}$ and the i-th summand is χ^{-i}-isotypic. In particular, the χ^{-1}-isotypic summand of $f_*\mathcal{O}_{X'}$ is $\mathcal{L}^{-1} \cdot \phi^{-1/r} \cong \mathcal{L}^{-1}$.

By duality for the finite morphism f, the trace pairing

$$f_*\mathcal{O}_{X'} \otimes f_*\omega_{X'} \to \omega_{X^\circ}$$

induces

$$f_*\omega_{X'} \cong \mathcal{H}om(f_*\mathcal{O}_{X'}, \omega_{X^\circ}).$$

The pairing is μ_r-invariant, so it vanishes on the product of the χ^i-isotypic summand of $f_*\omega_{X'}$ and the χ^{-j}-isotypic summand of $f_*\mathcal{O}_{X'}$ unless $\chi^i = \chi^j$. Thus the χ-isotypic summand of $f_*\omega_{X'}$ is

$$\mathcal{H}om(\mathcal{L}^{-1}, \omega_{X^\circ}) = \omega_{X^\circ} \otimes \mathcal{L}.$$

By Set-up 3.1(3),

$$\operatorname{mult}_{B_i} \operatorname{div}(\phi) = -r \operatorname{mult}_{B_i}(K_{X^\circ} + B^\circ),$$

so the ramification index of f over B_i is the denominator m_i of b_i, and f is étale over the generic point of any prime divisor not contained in $\operatorname{Supp} B^\circ$. In particular, no component of $(B^\circ)^{=1}$ is a branch divisor of f, and D' is reduced. Since $(B^\circ)^{=1}$ is Cartier, the projection formula implies that the χ-isotypic summand of $f_*\omega_{X'}(D')$ is

$$\omega_{X^\circ}((B^\circ)^{=1}) \otimes \mathcal{L} = \mathcal{O}_{X^\circ}(K_{X^\circ} + (B^\circ)^{=1} - K_{X^\circ} - \lfloor B^\circ \rfloor) = \mathcal{O}_{X^\circ}(G).$$

Next we show $h_*\omega_{X''}(E'') = \omega_{X'}(D')$. By the ramification formula, (X', D') is lc and X' is klt, so $h_*\omega_{X''}(E'') = \omega_{X'}(D')$ since E'' contains every h-exceptional prime divisor over $\operatorname{Supp} D'$.

For any $u \in Z^\circ$, we have [Del71, Section 3.2]

$$F^d H^d(X''_u \backslash E''_u, \mathbb{C}) = H^0(X''_u, \omega_{X''_u}(E''_u)).$$

Since π'' is smooth, the strata of E'' are smooth over Z°, and $\dim F^d H^d(X''_u \backslash E''_u, \mathbb{C})$ is independent of u (by the E_1-degeneration of [Del71, Corollaire 3.2.13(ii)], the numbers $h^q(X''_u, \Omega^p_{X''_u}\langle E''_u\rangle)$, $p+q=d$, for the sheaves of logarithmic forms of [Del71, (3.1.2)], are upper semicontinuous in u with constant sum $\dim H^d(X''_u \backslash E''_u, \mathbb{C})$), we have

$$F^d\mathbb{H} = \pi''_*\omega_{X''/Z^\circ}(E'') = \pi''_*\omega_{X''}(E'') \otimes \omega_{Z^\circ}^{-1}.$$

Since μ_r acts trivially on X° and

$$\pi''_*\omega_{X''}(E'') = \pi_* f_* h_* \omega_{X''}(E'') = \pi_* f_* \omega_{X'}(D'),$$

taking χ-isotypic parts implies

$$F^d\mathbb{H}_\chi \cong \pi_*\mathcal{O}_{X^\circ}(G) \otimes \omega_{Z^\circ}^{-1}.$$

Since $F^d\mathbb{H}_\chi$ is locally free, it remains to show $\operatorname{rank} \pi_*\mathcal{O}_{X^\circ}(G) = 1$. We show

$$\mathcal{O}_{X^\circ}(\lceil \mathbf{A}^*(X^\circ, B^\circ)\rceil) = \mathcal{O}_{X^\circ}(G),$$

where the left side is the sheaf of Definition 2.2. The coefficient g_i of G along B_i is $-\lfloor b_i \rfloor$ if $b_i < 1$ and 0 if $b_i = 1$. Thus $0 \le g_i < 1 - b_i$ if $b_i < 1$. On X° itself, $a(B_i, X^\circ, B^\circ) = 1 - b_i$, so

$$\lceil \mathbf{A}^*(X^\circ, B^\circ)_{X^\circ}\rceil = \sum_{b_i<1} \lceil -b_i \rceil B_i = G,$$

which implies $\subset$. Conversely, let $q : Y \to X^\circ$ be a projective birational morphism from a normal variety and E a prime divisor on Y. Since $\operatorname{Supp} B^\circ$ is snc, $a(E, X^\circ, \sum_i B_i) \geq 0$, hence

$$a(E, X^\circ, B^\circ) \geq \sum_i (1 - b_i) \operatorname{mult}_E q^* B_i \geq \sum_i g_i \operatorname{mult}_E q^* B_i = \operatorname{mult}_E q^* G,$$

and the second inequality is strict if $\operatorname{mult}_E q^* G > 0$. If $\operatorname{mult}_E q^* G > 0$, then

$$a(E, X^\circ, B^\circ) > \operatorname{mult}_E q^* G \geq 1,$$

so the coefficient of $\lceil \mathbf{A}^*(X^\circ, B^\circ)_Y \rceil$ along E is $\lceil a(E, X^\circ, B^\circ) - 1 \rceil \geq \operatorname{mult}_E q^* G$. If $\operatorname{mult}_E q^* G = 0$, then the coefficient of $\lceil \mathbf{A}^*(X^\circ, B^\circ)_Y \rceil$ along E is ≥ 0 since $a(E, X^\circ, B^\circ) \geq 0$. Thus $\lceil \mathbf{A}^*(X^\circ, B^\circ)_Y \rceil \geq q^* G$, and any rational function ψ with $\operatorname{div}(\psi) + G \geq 0$ satisfies

$$\operatorname{div}(q^* \psi) + \lceil \mathbf{A}^*(X^\circ, B^\circ)_Y \rceil \geq 0,$$

which implies $\supset$. Restricting Definition 2.4(2) to Z°, we obtain $\operatorname{rank} \pi_* \mathcal{O}_{X^\circ}(G) = 1$. □

In [BFMT25, Construction-Definition 6.23], the display for the χ-isotypic summand of the pushforward of logarithmic canonical forms omits the twist by $(B^\circ)^{=1}$, and the divisor there should be $-\lfloor B^{<0} \rfloor = G$, where $B^{<0}$ is the part of B° with negative coefficients. The formula of Lemma 3.2 is the one of [FG14, Remark 3.9].

Proposition 3.3 (cf. [BFMT25, Construction-Definition 6.23, Remark 6.25, and Theorem 6.28])**.** *Notation and conditions as in Set-up 3.1. Then the moduli* $\boldsymbol{b}$*-divisor* $\mathbf{M}$ *of* π *descends to* Z*, and*

$$\mathcal{O}_Z(mM_Z) \cong ((F^d \mathbb{H}_\chi)_Z)^{\otimes m}$$

for some positive integer m *such that* mM_Z *is integral.*

Proof. By Lemma 3.2 and the projection formula, the χ-isotypic summand of $f_* \omega_{X'}$ is $\omega_{X^\circ} \otimes \mathcal{L}$, and $F^d \mathbb{H}_\chi$ is a line bundle. Thus χ is the character fixed in [BFMT25, Construction-Definition 6.23] (the one whose isotypic summand of $f_* \omega_{X'}$ is $\omega_{X^\circ} \otimes \mathcal{L}$), and $\mathbb{H}_\chi$ gives the Hodge-theoretic moduli $\boldsymbol{b}$-divisor of π defined there. Let L_Z be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor with $K_X + B \sim_{\mathbb{Q}} \pi^* L_Z$ as in Definition 2.5, so that $M_Z = L_Z - K_Z - B_Z$, and let M^{H} be a divisor on Z with $\mathcal{O}_Z(M^{\mathrm{H}}) \cong (F^d \mathbb{H}_\chi)_Z$.

Let $g : Z' \to Z$ be a projective birational morphism from a normal variety Z', possibly $g = \mathrm{id}_Z$, in which case $\mathbf{M}_{Z'} = M_Z$. By [BFMT25, Construction-Definition 6.23 and Remark 6.25], the trace on Z' of the Hodge-theoretic moduli $\boldsymbol{b}$-divisor of π is linearly equivalent to $g^* M^{\mathrm{H}}$. Let $\pi_{Z'} : (X_{Z'}, B_{X_{Z'}}) \to Z'$ be the lc-trivial fibration over Z' in Definition 2.5, with the induced morphism $h_{Z'} : X_{Z'} \to X$, and let $N_{Z'}$ be its Shokurov moduli divisor [BFMT25, Definition 6.21] formed with $h_{Z'}^* \phi$, that is,

$$r(K_{X_{Z'}} + B_{X_{Z'}}) + \operatorname{div}(h_{Z'}^* \phi) = r \pi_{Z'}^* (K_{Z'} + \mathbf{B}_{Z'} + N_{Z'})$$

(the boundary divisor of [BFMT25, Definition 6.20] is the discriminant $\mathbf{B}_{Z'}$). Since $\pi_{Z'}$ is a contraction and $K_{X_{Z'}} + B_{X_{Z'}} \sim_{\mathbb{Q}} \pi_{Z'}^* g^* L_Z$, we have $K_{Z'} + \mathbf{B}_{Z'} + N_{Z'} \sim_{\mathbb{Q}} g^* L_Z$, that is,

$$N_{Z'} \sim_{\mathbb{Q}} g^* L_Z - K_{Z'} - \mathbf{B}_{Z'} = \mathbf{M}_{Z'}.$$

By [BFMT25, Theorem 6.28] for $\pi_{Z'}$, there exists a positive integer c with $cN_{Z'} \sim cg^* M^{\mathrm{H}}$. Thus

$$\mathbf{M}_{Z'} \sim_{\mathbb{Q}} g^* M^{\mathrm{H}}.$$

For $g = \mathrm{id}_Z$, we obtain $M_Z \sim_{\mathbb{Q}} M^{\mathrm{H}}$, so there exists a positive integer m such that mM_Z is integral and $mM_Z \sim mM^{\mathrm{H}}$, which is the second conclusion. For an arbitrary g, we obtain $\mathbf{M}_{Z'} - g^* M_Z \sim_{\mathbb{Q}} 0$, and $g_*(\mathbf{M}_{Z'} - g^* M_Z) = 0$ by Definition 2.2. By the negativity lemma [KM98, Lemma 3.39(1)], $\mathbf{M}_{Z'} = g^* M_Z$. Since Z is smooth, $\mathbf{M}$ descends to Z. □

Lemma 3.4. *Let U be a smooth variety and $\pi_U : X_U \to U$ a smooth projective contraction. Let B_{X_U} be a $\mathbb{Q}$-divisor on X_U such that $\operatorname{Supp} B_{X_U}$ is an snc divisor. Let X_η be the generic fiber of π_U, and let*

$$r := \min\{k \in \mathbb{N}^+ \mid k(K_{X_\eta} + B_{X_U}|_{X_\eta}) \sim 0\}.$$

Let ϕ be a rational function on X_U such that no component of $r(K_{X_U} + B_{X_U}) + \operatorname{div}(\phi)$ dominates U for a canonical divisor K_{X_U} of X_U. Then there exist a dense open subset $U_0 \subset U$, a finite étale morphism $g : U' \to U_0$ from a smooth variety U', and a smooth projective variety Z containing U' as an open subset with the following property. If $\pi : (X, B) \to Z$ is an lc-trivial fibration between normal projective varieties such that B is a $\mathbb{Q}$-divisor and $(\pi^{-1}(U'), B|_{\pi^{-1}(U')})$ is the base change of (X_U, B_{X_U}) by g, then the conditions of Set-up 3.1 hold for π, $\Sigma := Z \backslash U'$, the pullbacks of ϕ and K_{X_U}, and some μ_r-equivariant resolution h.

Proof. Let d be the relative dimension of π_U. We divide the proof into several steps.

Step 1. Since π_U is a smooth contraction, the fibers of π_U are irreducible, so any prime divisor on X_U not dominating U is $\pi_U^* P$ for a prime divisor P on U. Thus $r(K_{X_U} + B_{X_U}) + \operatorname{div}(\phi) = \pi_U^* Q$ for a $\mathbb{Q}$-divisor Q on U.

Step 2. Let $f_U : X'_U \to X_U$ be the normalization of the cyclic cover of X_U given by an r-th root of ϕ, with the induced action of μ_r, let $h_U : X''_U \to X'_U$ be a μ_r-equivariant log resolution of $(X'_U, f_U^{-1}(\operatorname{Supp} B^{=1}_{X_U}))$, the restriction over U of one given by [AW97, Theorem 0.1] for the normalization of the cyclic cover of a projective compactification of X_U given by an r-th root of ϕ, with the induced μ_r-action, and let E''_U be the reduced inverse image of $B^{=1}_{X_U}$ on X''_U. By generic smoothness and [Ver76, Théorèmes (2.2), (3.3), and (4.14)], since π_U and $X''_U \to U$ are projective, there exists a dense open subset $U_0 \subset U \setminus \operatorname{Supp} Q$ such that the morphism $X''_U \to U$ and each stratum of E''_U are smooth over U_0, and $(X_U, \operatorname{Supp} B_{X_U})$ and (X''_U, E''_U) are topologically locally trivial over U_0. Let X_{U_0}, X'_{U_0}, X''_{U_0}, and E''_{U_0} be the inverse images of U_0 in X_U, X'_U, X''_U, and E''_U, and let $\pi''_{U_0} : X''_{U_0} \to U_0$ be the induced morphism. Then

$$\mathbb{H}_{U_0} := R^d \pi''_{U_0 *} \mathbb{Z}_{X''_{U_0} \setminus E''_{U_0}}$$

is a local system on U_0.

Step 3. Fix $u \in U_0$, let Λ be the stalk of $\mathbb{H}_{U_0}$ at u, and let $\Lambda_{\mathrm{tors}} \subset \Lambda$ be the torsion subgroup. Let $g : U' \to U_0$ be the connected finite étale cover corresponding to the kernel of the action of $\pi_1(U_0, u)$ on the finite groups $\Lambda/3\Lambda$ and Λ_{tors}, and let Z be a smooth projective variety containing U' as an open subset such that $Z \backslash U'$ is an snc divisor. Let T be the local monodromy of $g^* \mathbb{H}_{U_0}$ around a component of $Z \backslash U'$. Then T acts trivially on $\Lambda/3\Lambda$ and on Λ_{tors}.

The local system $\mathbb{H}_{U_0} \otimes \mathbb{Q}$ has a filtration W by sub-local systems whose graded pieces are subquotients of the cohomology local systems of X''_{U_0} and of the intersections of components of E''_{U_0} over U_0 [Del71, (3.2.4.1)], and by **Step 2**, X''_{U_0} and the intersections of components of E''_{U_0} are smooth projective over U_0. The local monodromy of the cohomology of a smooth projective family is quasi-unipotent by Borel's lemma [Sch73, 4.5] (see [Kol07, p. 160]), so T is quasi-unipotent. Let ζ be an eigenvalue of T on $\Lambda \otimes \mathbb{C}$. Then ζ is a root of unity. Since T acts trivially on $\Lambda/3\Lambda$, $(T - \mathrm{id}_\Lambda)/3$ is an endomorphism of $\Lambda/\Lambda_{\mathrm{tors}}$, so $(\zeta - 1)/3$ is an algebraic integer. Any conjugate of $(\zeta - 1)/3$ is $(\zeta' - 1)/3$ for a root of unity ζ', hence has absolute value $\leq 2/3 < 1$. If $\zeta \neq 1$, then the norm of the non-zero algebraic integer $(\zeta - 1)/3$ is a non-zero integer of absolute value < 1, which is impossible. Thus $\zeta = 1$, and T is unipotent on $\Lambda/\Lambda_{\mathrm{tors}}$. Since $T - \mathrm{id}_\Lambda$ vanishes on Λ_{tors}, T is unipotent on Λ. The argument of **Step 3** is similar to one used for [BFMT25, Theorem 6.28].

Step 4. Let $\pi : (X, B) \to Z$ be an lc-trivial fibration between normal projective varieties such that B is a $\mathbb{Q}$-divisor and $(\pi^{-1}(U'), B|_{\pi^{-1}(U')})$ is the base change of (X_U, B_{X_U}) by g. Let $X^\circ := \pi^{-1}(U')$, $B^\circ := B|_{X^\circ}$, and let $X_{\eta'}$ be the generic fiber of π. In this step we show

$$\min\{n \in \mathbb{N}^+ \mid n(K_{X_{\eta'}} + B|_{X_{\eta'}}) \sim 0\} = r,$$

and that the base change of h_U by g is a μ_r-equivariant resolution of the normalization of the cyclic cover of X° given by an r-th root of ϕ.

Since π_U is a smooth contraction, X_η is geometrically integral, so by flat base change, for any finite field extension k' of $\mathbb{C}(U)$ and $X_{k'} := X_\eta \otimes_{\mathbb{C}(U)} k'$, a divisor on X_η is linearly equivalent to 0 if and only if its pullback to $X_{k'}$ is. Since $X_{\mathbb{C}(U')} = X_{\eta'}$, this proves the first display of **Step 4**.

We show that $\mathbb{C}(U)$ is algebraically closed in $\mathbb{C}(X_U)(\phi^{1/r})$. Let k' be the algebraic closure of $\mathbb{C}(U)$ in $\mathbb{C}(X_U)(\phi^{1/r})$. Then k' is a finite extension of $\mathbb{C}(U)$. By Kummer theory, the intermediate field $\mathbb{C}(X_U)k'$ of $\mathbb{C}(X_U)(\phi^{1/r})/\mathbb{C}(X_U)$ is $\mathbb{C}(X_U)(\phi^{1/m})$ for a positive integer m dividing r, where $\phi^{1/m} := (\phi^{1/r})^{r/m}$. Since X_η is geometrically integral, $\mathbb{C}(X_U)k'$ is the function field of $X_{k'}$, and by **Step 1**,

$$\frac{r}{m}\big(K_{X_\eta} + B_{X_U}|_{X_\eta}\big) + \operatorname{div}(\phi^{1/m}) = 0$$

on $X_{k'}$. By the previous paragraph, $(r/m)(K_{X_\eta} + B_{X_U}|_{X_\eta}) \sim 0$ on X_η, so $m = 1$ by the minimality of r. Thus $k' \subset \mathbb{C}(X_U)$, and $k' = \mathbb{C}(U)$.

Let X'° and X''° be the base changes of X'_{U_0} and X''_{U_0} by g, and let $h : X''^\circ \to X'^\circ$ be the base change of h_U. Since g is étale and $\mathbb{C}(U)$ is algebraically closed in $\mathbb{C}(X_U)(\phi^{1/r})$, X'° and X''° are normal and irreducible, with function field

$$\mathbb{C}(X_U)(\phi^{1/r}) \otimes_{\mathbb{C}(U)} \mathbb{C}(U') = \mathbb{C}(X^\circ)(\phi^{1/r}).$$

Thus $X'^\circ \to X^\circ$ is the normalization of the cyclic cover of X° given by an r-th root of ϕ, with the induced action of μ_r, and h is a μ_r-equivariant resolution.

Step 5. We verify Set-up 3.1(1)–(5) for π, $\Sigma = Z\backslash U'$, the pullbacks of ϕ and K_{X_U}, and h. Condition (1) holds by assumption and since Z is smooth by **Step 3**. The relative dimension of π is d. By **Step 3**, $\Sigma = Z\backslash U'$ is an snc divisor. Since $X^\circ \to X_{U_0}$ is étale, X° is smooth, $\operatorname{Supp} B^\circ$ is an snc divisor, and the pullback of K_{X_U} is a canonical divisor of X°. By **Step 2**, $(X^\circ, \operatorname{Supp} B^\circ)$ is topologically locally trivial over U'. This is (2). Condition (3) follows from **Step 4** and from the equality of **Step 1** over $U_0 \subset U \setminus \operatorname{Supp} Q$, pulled back by the étale morphism $X^\circ \to X_{U_0}$. By **Steps 2 and 4**, h satisfies (4): the induced morphism $X''^\circ \to U'$ is smooth, and the reduced inverse image of $(B^\circ)^{=1}$ on X''° is the base change of E''_{U_0}, an snc divisor each of whose strata is smooth over U'. Finally, for the induced morphism $\pi'' : X''^\circ \to U'$ and the reduced inverse image E'' of $(B^\circ)^{=1}$, we have

$$R^d\pi''_*\mathbb{Z}_{X''^\circ\setminus E''} = g^*\mathbb{H}_{U_0},$$

so (5) follows from **Step 3**. □

Lemma 3.5. *Let $\pi : X \to Z$ be a contraction between normal projective varieties with X $\mathbb{Q}$-factorial, B^h a $\mathbb{Q}$-divisor on X, r a positive integer, and ϕ a rational function on X such that no component of $r(K_X + B^h) + \operatorname{div}(\phi)$ dominates Z for a canonical divisor K_X of X, and let $B^v := -\frac{1}{r}(r(K_X + B^h) + \operatorname{div}(\phi))$. Assume that (X, B^h) is sub-lc over the generic point of Z and $\operatorname{rank} \pi_*\mathcal{O}_X(\lceil \mathbf{A}^*(X, B^h)\rceil) = 1$. Then $\pi : (X, B^h + B^v) \to Z$ is an lc-trivial fibration with $K_X + B^h + B^v \sim_{\mathbb{Q}} 0$.*

Proof. By the definition of B^v, $r(K_X + B^h + B^v) + \operatorname{div}(\phi) = 0$, and $(X, B^h + B^v)$ coincides with (X, B^h) over a non-empty open subset of Z. □

4. Period map and monodromy of cyclic covers of elliptic curves

In this section, we construct a family $\mathcal{C} \to U$ of cyclic covers of degree 5 of curves of genus 1 over a smooth base U of dimension 5, and we study the period map and the monodromy group of $\mathcal{C} \to U$ (Proposition 4.2).

Construction-Proposition 4.1. Let χ be the character $\zeta \to \zeta^{-1}$ of μ_5. Then there exist a smooth variety U of dimension 5, a smooth projective morphism $e : \mathcal{E} \to U$ with a section $P \subset \mathcal{E}$, a reduced divisor $D \subset \mathcal{E}$ finite and étale of degree 5 over U with $D \cap P = \emptyset$, and a μ_5-cover

$c : \mathcal{C} \to \mathcal{E}$ with $e \circ c$ smooth and projective, satisfying the following, where E_u, P_u, D_u, and C_u are the fibers over $u \in U$.

(1) E_u is a curve of genus 1, $D_u \sim 5P_u$, and $c_u : C_u \to E_u$ is the cyclic cover of degree 5 branched along D_u with $c_{u*}\mathcal{O}_{C_u} \cong \bigoplus_{i=0}^{4} \mathcal{O}_{E_u}(-iP_u)$, where μ_5 acts on $\mathcal{O}_{E_u}(-iP_u)$ by χ^{-i}.
(2) For any $u \in U$, the Kodaira–Spencer map $T_{U,u} \to H^1(E_u, T_{E_u}(-D_u))$ of the family $(\mathcal{E}, D) \to U$ at u is an isomorphism.
(3) The kernel $\mathbb{V}$ of the trace map
$$R^1(e \circ c)_*\mathbb{Q}_{\mathcal{C}} \to R^1 e_* \mathbb{Q}_{\mathcal{E}}$$
is a polarized variation of $\mathbb{Q}$-Hodge structure of weight 1 and rank 20 with an action of $\mathbb{Q}(\zeta_5)$, where ζ_5 is a primitive fifth root of unity, and
$$\mathbb{V} \otimes \mathbb{C} = \bigoplus_{i=1}^{4} \mathbb{V}_{\chi^i}$$
for the χ^i-isotypic parts $\mathbb{V}_{\chi^i}$ of $\mathbb{V} \otimes \mathbb{C}$, with $\operatorname{rank} \mathbb{V}_{\chi^i} = 5$ and $\mathbb{V}^{1,0}_{\chi^i,u} = H^0(E_u, \omega_{E_u}(iP_u))$, where
$$\mathbb{V}^{p,q}_{\chi^i,u} := \mathbb{V}^{p,q}_u \cap \mathbb{V}_{\chi^i,u}$$
and $H^0(E_u, \omega_{E_u}(iP_u))$ is embedded into $H^0(C_u, \omega_{C_u})$ by $\eta \to c_u^*\eta \cdot (\xi^{1/5}|_{C_u})^{-i}$ for the rational function $\xi^{1/5}$ on $\mathcal{C}$ in (5).
(4) $\mathcal{A} := \mathbb{V}^{1,0}_{\chi}$ is a line subbundle of $\mathbb{V}_{\chi} \otimes \mathcal{O}_U$.
(5) c is the normalization of $\mathcal{E}$ in $\mathbb{C}(\mathcal{E})(\xi^{1/5})$ for a rational function ξ on $\mathcal{E}$ with $\operatorname{div}(\xi) = D - 5P$, and μ_5 acts on $\xi^{1/5}$ by χ^{-1}.

Proof. Let $\mathcal{M}$ be the moduli stack of pairs (E, D), where E is a smooth projective curve of genus 1 and $D \geq 0$ is a reduced divisor of degree 5 on E. The tangent space of $\mathcal{M}$ at (E, D) is $H^1(E, T_E(-D))$ [ACG11, Chapter XI, Section 2 and Theorem (2.12)]. Since $\deg T_E(-D) = -5$, by the Riemann–Roch theorem,
$$h^0(E, T_E(-D)) = 0 \quad \text{and} \quad h^1(E, T_E(-D)) = 5.$$
Since $h^0(E, T_E(-D)) = 0$ and the obstruction space $H^2(E, T_E(-D))$ vanishes, $\mathcal{M}$ is a smooth Deligne–Mumford stack with $\dim \mathcal{M} = h^1(E, T_E(-D)) = 5$.

Let $\mathcal{R}$ be the moduli stack of quadruples (E, L, D, α), where (E, D) is a pair parametrized by $\mathcal{M}$, L is a line bundle of degree 1 on E, and $\alpha : L^{\otimes 5} \to \mathcal{O}_E(D)$ is an isomorphism. Isomorphisms of quadruples include isomorphisms $E \to E'$ of the curves. The fiber of $\mathcal{R} \to \mathcal{M}$ over (E, D) is $\{L \mid L^{\otimes 5} \cong \mathcal{O}_E(D)\}$, a torsor under $\operatorname{Pic}^0(E)[5]$, and α is unique up to the isomorphisms $\lambda^{1/5} \operatorname{id}_L$ for $\lambda \in \mathbb{C}^*$. Thus $\mathcal{R} \to \mathcal{M}$ is étale. By the Riemann–Roch theorem, $L \cong \mathcal{O}_E(P)$ for a unique point $P \in E$, and $D \in |5P| \cong \mathbb{P}^4$. Since $|5P|$ is base point free, the divisors in $|5P|$ containing P form a hyperplane of $|5P|$. Let $\mathcal{Z} \subset \mathcal{R}$ be the locus where $P \in \operatorname{Supp} D$. Then
$$\dim \mathcal{Z} \leq \dim \mathcal{M}_{1,1} + \dim\{D \in |5P| \mid P \in \operatorname{Supp} D\} = 1 + 3 = 4,$$
where $\mathcal{M}_{1,1}$ is the moduli stack of curves of genus 1 with a point.

To (E, L, D, α) we attach the cyclic cover
$$c_E : C := \operatorname{Spec}_E \bigoplus_{i=0}^{4} L^{\otimes -i} \to E,$$
where, for $i + j \geq 5$, the multiplication is the composition
$$L^{\otimes -i} \otimes L^{\otimes -j} \to L^{\otimes -(i+j)} \cong L^{\otimes -(i+j-5)}(-D) \subset L^{\otimes -(i+j-5)},$$
the isomorphism being induced by α, and $\zeta \in \mu_5$ acts on $L^{\otimes -i}$ by ζ^i [EV92, 3.5 and 3.9]. Thus μ_5 acts on $L^{\otimes -i}$ by χ^{-i}. Near a point of D with local coordinate x on E with $D = (x = 0)$,

the curve C is given by $y^5 = x$, since D is reduced. Thus C is smooth, c_E is totally ramified over D and étale elsewhere, and C is connected, because any connected component of C maps onto E and the fiber of c_E over a point of D is one point. Since $\deg D = 5 \deg L = 5$, by the Riemann–Hurwitz formula,

$$2g(C) - 2 = 5(2g(E) - 2) + 4 \deg D = 20, \quad \text{so} \quad g(C) = 11.$$

A level-3 structure on (E, L, D, α) is an isomorphism $H^1(C, \mathbb{Z}/3) \cong (\mathbb{Z}/3)^{22}$. An automorphism of (E, L, D, α) induces an automorphism of C, and the induced automorphism determines the automorphism of the quadruple, since it determines the automorphism of $E = C/\mu_5$ and of the summand $L^{\otimes -1}$ of $c_{E*}\mathcal{O}_C$. Since $g(C) = 11 \geq 2$, an automorphism of C acting trivially on $H^1(C, \mathbb{Z}/3)$ is the identity (Serre, [ACG11, Chapter XVI, Proposition (2.8)]). Thus the moduli stack $\mathcal{R}^{(3)}$ of quadruples with a level-3 structure has trivial automorphism groups, so $\mathcal{R}^{(3)}$ is an algebraic space [Stacks, Tag 04SZ], finite and étale over $\mathcal{R}$, and hence étale over $\mathcal{M}$. Let U_0 be a connected component of a dense open subscheme [Stacks, Tag 06NH] of $\mathcal{R}^{(3)}$. It is a smooth variety with $\dim U_0 = \dim \mathcal{M} = 5$. Let $U \subset U_0$ be the open subset of points u such that $P_u \notin \operatorname{Supp} D_u$, where $L_u \cong \mathcal{O}_{E_u}(P_u)$ for the quadruple $(E_u, L_u, D_u, \alpha_u)$ at u. Since $U_0 \to \mathcal{R}$ is quasi-finite and $\dim \mathcal{Z} \leq 4$, U is dense in U_0.

Let $(\mathcal{E}, \mathcal{L}, D, \alpha)$ be the universal quadruple over U, $e : \mathcal{E} \to U$ the family of curves, and

$$c : \mathcal{C} := \operatorname{Spec}_{\mathcal{E}} \bigoplus_{i=0}^{4} \mathcal{L}^{\otimes -i} \to \mathcal{E}$$

the cyclic cover attached to $(\mathcal{E}, \mathcal{L}, D, \alpha)$, whose multiplication and μ_5-action are those of c_E with $\mathcal{L}$ in place of L. Since $h^0(E_u, \mathcal{L}|_{E_u}) = 1$ and $h^1(E_u, \mathcal{L}|_{E_u}) = 0$ for any $u \in U$, $e_*\mathcal{L}$ is a line bundle. Possibly shrinking U, we may assume that $e_*\mathcal{L} \cong \mathcal{O}_U$. A generator σ of $e_*\mathcal{L}$ is a section of $\mathcal{L}$ whose zero divisor P satisfies $P \cap E_u = P_u$ for any $u \in U$. Thus P is a section of e, $\mathcal{L} \cong \mathcal{O}_{\mathcal{E}}(P)$ via σ, and $D \cap P = \emptyset$ by the definition of U. The divisor D is finite and flat of degree 5 over U with reduced fibers, so D is étale over U. Near D, the variety $\mathcal{C}$ is given by $y^5 = x$ for a local equation x of D, and c is étale over $\mathcal{E} \backslash D$. Since D is étale over U, $\mathcal{C}$ is smooth over U, so $e \circ c$ is smooth and projective.

(1) The restriction of c to E_u is the cover c_{E_u} attached to $(E_u, \mathcal{L}|_{E_u}, D_u, \alpha|_{E_u})$, and

$$\mathcal{L}|_{E_u} \cong \mathcal{O}_{E_u}(P_u).$$

Thus (1) follows from the properties of c_E above, and $D_u \sim 5P_u$ via α.

(2) The Kodaira–Spencer map of $(\mathcal{E}, D)$ at u is the tangent map at u of the classifying morphism $U \to \mathcal{M}$. The classifying morphism is the composition of the open immersion $U \to \mathcal{R}^{(3)}$ and the étale morphisms $\mathcal{R}^{(3)} \to \mathcal{R} \to \mathcal{M}$, so the tangent map is an isomorphism.

(5) Via σ, α is an isomorphism $\mathcal{O}_{\mathcal{E}}(5P) \to \mathcal{O}_{\mathcal{E}}(D)$ of subsheaves of the sheaf of rational functions on $\mathcal{E}$, so α is the multiplication by ξ^{-1} for a rational function ξ on $\mathcal{E}$ with $\operatorname{div}(\xi) = D - 5P$. Since $D \neq 0$ is reduced, ξ is not the fifth power of a rational function on $\mathcal{E}$, so $[\mathbb{C}(\mathcal{E})(\xi^{1/5}) : \mathbb{C}(\mathcal{E})] = 5$. The map

$$\bigoplus_{i=0}^{4} \mathcal{O}_{\mathcal{E}}(-iP) \to \mathbb{C}(\mathcal{E})(\xi^{1/5}), \quad (s_i)_{i=0}^4 \to \sum_{i=0}^{4} s_i \xi^{i/5},$$

is injective, and it is a homomorphism of $\mathcal{O}_{\mathcal{E}}$-algebras, since $s\xi^{i/5} \cdot s'\xi^{j/5} = (ss'\xi) \cdot \xi^{(i+j-5)/5}$ for $i + j \geq 5$. Thus $\mathcal{C}$ is finite over $\mathcal{E}$ with function field $\mathbb{C}(\mathcal{E})(\xi^{1/5})$. Since $\mathcal{C}$ is smooth, $\mathcal{C}$ is the normalization of $\mathcal{E}$ in $\mathbb{C}(\mathcal{E})(\xi^{1/5})$. The summand $\mathcal{O}_{\mathcal{E}}(-P)$, on which μ_5 acts by χ^{-1}, is $\mathcal{O}_{\mathcal{E}}(-P) \cdot \xi^{1/5}$, so μ_5 acts on $\xi^{1/5}$ by χ^{-1}.

(3) Since $e \circ c$ is smooth and projective, $R^1(e \circ c)_*\mathbb{Q}_{\mathcal{C}}$ is a polarized variation of $\mathbb{Q}$-Hodge structure of weight 1, and the trace map is a morphism of variations, so $\mathbb{V}$ is a polarized

sub-variation. For any $u \in U$, the trace c_{u*} and the pullback c_u^* satisfy

$$c_{u*} \circ c_u^* = 5 \quad \text{and} \quad c_u^* \circ c_{u*} = \sum_{\zeta \in \mu_5} \zeta^*,$$

so $\mathbb{V}_u$ is the kernel of $\sum_{\zeta\in\mu_5} \zeta^*$ on $H^1(C_u, \mathbb{Q})$. Thus

$$\mathbb{V} \otimes \mathbb{C} = \bigoplus_{i=1}^{4} \mathbb{V}_{\chi^i},$$

and the group algebra $\mathbb{Q}[\mu_5]$ acts on $\mathbb{V}$ through

$$\mathbb{Q}[\mu_5]/\Big(\sum_{\zeta\in\mu_5} \zeta\Big) \cong \mathbb{Q}(\zeta_5).$$

Let $u \in U$, $\xi_u := \xi|_{E_u}$, and $\xi_u^{-i/5} := (\xi^{1/5}|_{C_u})^{-i}$ for $0 \le i \le 4$. Then $\operatorname{div}(\xi_u) = D_u - 5P_u$. Let $\widetilde{D}_u := c_u^{-1}(D_u)_{\mathrm{red}}$, so $c_u^* D_u = 5\widetilde{D}_u$ and the ramification divisor of c_u is $4\widetilde{D}_u$. For $0 \le i \le 4$ and $\eta \in H^0(E_u, \omega_{E_u}(iP_u))$, we have

$$\operatorname{div}(c_u^*\eta \cdot \xi_u^{-i/5}) = c_u^*(\operatorname{div}(\eta) + iP_u) + (4-i)\widetilde{D}_u \ge 0,$$

and μ_5 acts on $\xi_u^{-i/5}$ by χ^i. Thus

$$H^0(E_u, \omega_{E_u}(iP_u)) \to H^0(C_u, \omega_{C_u})_{\chi^i}, \quad \eta \to c_u^*\eta \cdot \xi_u^{-i/5},$$

is an injective linear map for $0 \le i \le 4$. Since

$$\sum_{i=0}^{4} h^0(E_u, \omega_{E_u}(iP_u)) = 1+1+2+3+4 = 11 = g(C_u) = h^0(C_u, \omega_{C_u}),$$

all five maps are isomorphisms. Thus, for $1 \le i \le 4$,

$$\mathbb{V}^{1,0}_{\chi^i,u} = H^0(C_u, \omega_{C_u})_{\chi^i} \cong H^0(E_u, \omega_{E_u}(iP_u)) \quad \text{and} \quad \dim \mathbb{V}^{1,0}_{\chi^i,u} = i.$$

Since μ_5 acts on $\mathbb{V}_u$ by $\mathbb{Q}$-linear maps, complex conjugation maps $\mathbb{V}_{\chi^i,u}$ onto $\mathbb{V}_{\chi^{-i},u}$, so

$$\dim \mathbb{V}^{0,1}_{\chi^i,u} = \dim \mathbb{V}^{1,0}_{\chi^{5-i},u} = 5 - i.$$

Thus $\operatorname{rank} \mathbb{V}_{\chi^i} = 5$ for $1 \le i \le 4$ and $\operatorname{rank} \mathbb{V} = 20$.

(4) By (3), $\dim \mathbb{V}^{1,0}_{\chi,u} = 1$ for any $u \in U$. Since $\mathbb{V}^{1,0}$ is a holomorphic subbundle of $\mathbb{V} \otimes \mathcal{O}_U$ stable under μ_5, its image $\mathbb{V}^{1,0}_\chi$ under the projector $\frac{1}{5}\sum_{\zeta\in\mu_5} \chi(\zeta)^{-1}\zeta^*$ onto $\mathbb{V}_\chi \otimes \mathcal{O}_U$ is a line subbundle of $\mathbb{V}_\chi \otimes \mathcal{O}_U$. □

Proposition 4.2. *Notation as in Construction-Proposition 4.1. Let $g : U' \to U$ be a dominant generically finite morphism from a smooth variety U'. Then the following hold.*

(1) For a general point $u' \in U'$, the map

$$T_{U',u'} \to \operatorname{Hom}(\mathbb{V}^{1,0}_{g(u')}, \mathbb{V}^{0,1}_{g(u')})$$

given by the Gauss–Manin connection of $g^*\mathbb{V}$ is injective.

(2) For any $u' \in U'$, the identity component of the Zariski closure in $GL(\mathbb{V}_{g(u')})$ of the monodromy group of $g^*\mathbb{V}$ acts irreducibly on the $\mathbb{Q}$-vector space $\mathbb{V}_{g(u')}$.

Proof. (1) Since g is dominant and generically finite, $\dim U' = \dim U = 5$, and g is étale at a general point $u' \in U'$ by generic smoothness. At such u', the map in (1) is the composition of the isomorphism $dg : T_{U',u'} \to T_{U,g(u')}$ and the map

$$\overline{\nabla}_u : T_{U,u} \to \operatorname{Hom}(\mathbb{V}^{1,0}_u, \mathbb{V}^{0,1}_u)$$

given by the Gauss–Manin connection of $\mathbb{V}$ at $u = g(u')$. Thus it suffices to prove that $\overline{\nabla}_u$ is injective for any $u \in U$.

Fix $u \in U$ and write E, P, D, $\widetilde{D}$, C, c, ξ for E_u, P_u, D_u, $\widetilde{D}_u$, C_u, c_u, $\xi|_{E_u}$, and $\xi^{-j/5}$ for $\xi_u^{-j/5}$. Let

$$\kappa : T_{U,u} \to H^1(E, T_E(-D))$$

be the Kodaira–Spencer map of $(\mathcal{E}, D)$, an isomorphism by Construction-Proposition 4.1(2), and let $\kappa_{\mathcal{C}} : T_{U,u} \to H^1(C, T_C)$ be the Kodaira–Spencer map of $e \circ c$. By Griffiths' theorem [Voi02, Theorem 10.21], for $v \in T_{U,u}$, $\overline{\nabla}_u(v)$ is the restriction to the subspace $\mathbb{V}_u^{1,0}$ of $H^0(C, \omega_C)$ of the cup product

$$\kappa_{\mathcal{C}}(v) \cup - : H^0(C, \omega_C) \to H^1(C, \mathcal{O}_C).$$

Near a point of D, the curve C is given by $y^5 = x$ for a local coordinate x on E with $D = (x = 0)$, and the μ_5-invariant vector fields on C are the multiples of $y\partial_y = 5x\partial_x$ by invariant functions. Away from D, c is étale. Thus $(c_* T_C)^{\mu_5} = T_E(-D)$, and since c is finite and taking μ_5-invariants is exact,

$$\iota : H^1(E, T_E(-D)) = H^1(E, (c_* T_C)^{\mu_5}) \xrightarrow{\cong} H^1(C, T_C)^{\mu_5}.$$

By the local description of c above, a vector field on an open subset of E vanishing on D lifts uniquely to a μ_5-invariant vector field on its preimage in C. Thus the first-order deformation of C along v is obtained by gluing with the lifts of the gluing vector fields of the first-order deformation of (E, D), and $\kappa_{\mathcal{C}} = \iota \circ \kappa$.

Let $\mathrm{tr}_C : H^1(C, \omega_C) \to \mathbb{C}$ be the trace map. For $\alpha \in H^0(C, \omega_C)_{\chi^2}$ and $\beta \in H^0(C, \omega_C)_{\chi^3}$, we have

$$\mathrm{tr}_C\left((\kappa_{\mathcal{C}}(v) \cup \alpha) \cup \beta\right) = \mathrm{tr}_C\left(\kappa_{\mathcal{C}}(v) \cup (\alpha\beta)\right),$$

where $\alpha\beta \in H^0(C, \omega_C^{\otimes 2})^{\mu_5}$ since $\chi^2\chi^3 = 1$. Since the Serre duality pairing between $H^1(C, T_C)$ and $H^0(C, \omega_C^{\otimes 2})$ is μ_5-invariant, it restricts to a perfect pairing between $H^1(C, T_C)^{\mu_5}$ and $H^0(C, \omega_C^{\otimes 2})^{\mu_5}$. Let $v \in \operatorname{Ker} \overline{\nabla}_u$. Since $\mathbb{V}^{1,0}_{\chi^2,u} = H^0(C, \omega_C)_{\chi^2}$, we have $\kappa_{\mathcal{C}}(v) \cup \alpha = 0$ for any $\alpha \in H^0(C, \omega_C)_{\chi^2}$. Thus, if

$$H^0(C, \omega_C^{\otimes 2})^{\mu_5} = \mathrm{Span}\{\alpha\beta \mid \alpha \in H^0(C, \omega_C)_{\chi^2},\ \beta \in H^0(C, \omega_C)_{\chi^3}\}, \tag{4.1}$$

then $\kappa_{\mathcal{C}}(v) = 0$, so $v = 0$ since $\iota \circ \kappa$ is injective (compare [LO11, Proposition 4.1], for base curves of genus ≥ 2).

We prove (4.1). By Construction-Proposition 4.1(3), any

$$\alpha \in H^0(C, \omega_C)_{\chi^2} \quad \text{and} \quad \beta \in H^0(C, \omega_C)_{\chi^3}$$

are $\alpha = c^*\eta_2 \cdot \xi^{-2/5}$ and $\beta = c^*\eta_3 \cdot \xi^{-3/5}$ with $\eta_j \in H^0(E, \omega_E(jP))$, so

$$\alpha\beta = c^*(\eta_2\eta_3\xi^{-1}), \quad \text{where} \quad \eta_2\eta_3\xi^{-1} \in H^0(E, \omega_E^{\otimes 2}(D))$$

since $\mathrm{div}(\xi^{-1}) = 5P - D$. For $q \in H^0(E, \omega_E^{\otimes 2}(D))$,

$$\mathrm{div}(c^*q) = c^*(\mathrm{div}(q) + D) + 3\widetilde{D} \geq 0,$$

so the map

$$c^* : H^0(E, \omega_E^{\otimes 2}(D)) \to H^0(C, \omega_C^{\otimes 2})^{\mu_5}$$

is injective. By the Riemann–Roch theorem and Serre duality,

$$h^0(E, \omega_E^{\otimes 2}(D)) = 5 = h^1(E, T_E(-D)) = \dim H^1(C, T_C)^{\mu_5} = \dim H^0(C, \omega_C^{\otimes 2})^{\mu_5},$$

so c^* is an isomorphism. Let $L := \mathcal{O}_E(P)$. Since $\omega_E \cong \mathcal{O}_E$ and multiplication by ξ^{-1} is an isomorphism

$$H^0(E, \omega_E^{\otimes 2}(5P)) \to H^0(E, \omega_E^{\otimes 2}(D)),$$

(4.1) follows from the surjectivity of the multiplication map

$$H^0(E, L^{\otimes 2}) \otimes H^0(E, L^{\otimes 3}) \to H^0(E, L^{\otimes 5}).$$

Since $L^{\otimes 2}$ is base point free with $h^0(E, L^{\otimes 2}) = 2$ and $H^1(E, L) = 0$, the multiplication map is surjective by the base-point-free pencil trick [ACGH85, p. 126]. This proves (1).

(2) Let $u \in U$, $V := \mathbb{V}_u$, $\Gamma \subset GL(V)$ the monodromy group of $\mathbb{V}$ at u, that is, the image of $\pi_1(U,u)$, and $\overline{\Gamma}$ the Zariski closure of Γ in $GL(V)$. We call the identity component $\overline{\Gamma}^0$ the *connected algebraic monodromy group* of $\mathbb{V}$ at u. We first prove that $\overline{\Gamma}^0$ acts irreducibly on V.

We show that $\overline{\Gamma}^0$ contains the group G defined below. Write $D_u = x_1 + \cdots + x_5$. By [Rai25, Theorem 9.2] (on μ_N-covers of a curve of genus g with $n+2-2g$ ramification points for $n \geq \max\{5, 2g-1\}$), applied with $N=5$, $g=1$, and $n=5$ in the notation there, the family of μ_5-covers of E_u containing c_u in which the branch point x_1 moves in $E_u \backslash \{x_2, \ldots, x_5\}$ and $x_2, \ldots, x_5$ are fixed has large monodromy in the sense of [Rai25, Definition 3.1]. Let b_0 be a point of the base T_0 of the family in [Rai25, Theorem 9.2] over which the cover is c_u, and let T be the connected component of T_0 containing b_0, a smooth curve finite étale over a dense open subset of E_u (see the beginning of [Rai25, Section 9]). Since 5 is prime, the intermediate covers of c_u are C_u and E_u, so the Prym of [Rai25, (1.3)] is isogenous to the kernel of the trace. Thus the Prym local system on T has fiber V at b_0, with the lattice $V_{\mathbb{Z}} := V \cap H^1(C_u, \mathbb{Z})$, which is preserved by the monodromy, and we let $\Gamma_T \subset GL(V)$ be the monodromy group at b_0. Let $\langle \cdot, \cdot \rangle$ be the non-degenerate anti-Hermitian form on the $\mathbb{Q}(\zeta_5)$-vector space V, with respect to $\mathbb{Q}(\zeta_5)/\mathbb{Q}(\sqrt{5})$, preserved by Γ_T in [Rai25, Definition 3.1], and let

$$G := \mathrm{Res}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}\, SU(V, \langle \cdot, \cdot \rangle) \subset GL(V),$$

a connected algebraic group over $\mathbb{Q}$. Let S be the set of finite places $\mathfrak{p}$ of $\mathbb{Q}(\sqrt{5})$ of [Rai25, Definition 3.1]. Only finitely many finite places are not in S. Since $V_{\mathbb{Z}}$ and the Γ_T-stable lattice of [Rai25, Definition 3.1] are commensurable, removing finitely many places from S, we may assume that their completions at $\mathfrak{p}$ coincide for any $\mathfrak{p} \in S$. For $\mathfrak{p} \in S$, let $SU(V_{\mathbb{Z},\mathfrak{p}})$ be the stabilizer of the completion $V_{\mathbb{Z},\mathfrak{p}}$ of $V_{\mathbb{Z}}$ at $\mathfrak{p}$ in the group

$$SU(V, \langle \cdot, \cdot \rangle)(\mathbb{Q}(\sqrt{5})_{\mathfrak{p}}).$$

Largeness means that $\Gamma_T^1 := \Gamma_T \cap G(\mathbb{Q})$ has dense image in $\prod_{\mathfrak{p} \in S} SU(V_{\mathbb{Z},\mathfrak{p}})$.

Let ℓ be a prime number that splits in $\mathbb{Q}(\sqrt{5})$ as

$$\ell \mathcal{O}_{\mathbb{Q}(\sqrt{5})} = \mathfrak{p}_1 \mathfrak{p}_2 \quad \text{with} \quad \mathfrak{p}_1, \mathfrak{p}_2 \in S.$$

Such ℓ exists, since there exist infinitely many primes $\ell \equiv \pm 1 \bmod 5$, each of them splits in $\mathbb{Q}(\sqrt{5})$, and only finitely many finite places are not in S. Then

$$\mathbb{Q}(\sqrt{5}) \otimes \mathbb{Q}_\ell = \mathbb{Q}(\sqrt{5})_{\mathfrak{p}_1} \times \mathbb{Q}(\sqrt{5})_{\mathfrak{p}_2},$$

so the group

$$SU(V_{\mathbb{Z},\mathfrak{p}_1}) \times SU(V_{\mathbb{Z},\mathfrak{p}_2})$$

is an open subgroup of $G(\mathbb{Q}_\ell)$. It contains Γ_T^1 as a dense subgroup, since Γ_T^1 preserves $V_{\mathbb{Z}}$ and has dense image in $\prod_{\mathfrak{p} \in S} SU(V_{\mathbb{Z},\mathfrak{p}})$. Let $\overline{\Gamma_T^1}$ be the Zariski closure of Γ_T^1 in $GL(V)$. Since $\overline{\Gamma_T^1}(\mathbb{Q}_\ell)$ is closed in $GL(V \otimes \mathbb{Q}_\ell)$ for the ℓ-adic topology, $(\overline{\Gamma_T^1} \cap G)(\mathbb{Q}_\ell)$ contains an open subgroup of $G(\mathbb{Q}_\ell)$. Thus

$$\mathrm{Lie}(\overline{\Gamma_T^1} \cap G) \otimes \mathbb{Q}_\ell = \mathrm{Lie}(G) \otimes \mathbb{Q}_\ell,$$

so $\dim(\overline{\Gamma_T^1} \cap G) = \dim G$, and $G \subset (\overline{\Gamma_T^1})^0$ since G is connected. Thus $G \subset \overline{\Gamma_T}^0$.

Let $T' \to T$ be a connected finite étale cover over which the $\mathrm{Pic}^0(E_u)[5]$-torsor of the roots L and the $GL_{22}(\mathbb{Z}/3)$-torsor of level-3 structures are trivial, let $b_0' \in T'$ be a point over b_0, and let $T'^\circ \subset T'$ be a dense open subset containing b_0' over which an isomorphism α exists. Then the pullback of the family to T'° is the cyclic cover attached to a family of quadruples with a level-3 structure over T'°, equal at b_0' to the quadruple and the level-3 structure of u. Since T'° is connected, the classifying morphism $T'^\circ \to \mathcal{R}^{(3)}$ maps T'° into the connected component of $\mathcal{R}^{(3)}$ containing u. Since U is open in $\mathcal{R}^{(3)}$ and contains u, possibly removing finitely many points from T'°, we may assume that it is a morphism $T'^\circ \to U$. The pullback of $\mathbb{V}$ by $T'^\circ \to U$ is the Prym local system, so the image $\Gamma_{T'}$ of $\pi_1(T'^\circ, b_0')$ in $GL(V)$ is contained in Γ. Since

$\pi_1(T'^\circ, b_0') \to \pi_1(T', b_0')$ is surjective [Kol95, Proposition 2.10] and the image of $\pi_1(T', b_0')$ in $\pi_1(T, b_0)$ has finite index, $\Gamma_{T'}$ has finite index in Γ_T, so

$$G \subset \overline{\Gamma_T}^0 = \overline{\Gamma_{T'}}^0 \subset \overline{\Gamma}^0.$$

We show that G acts irreducibly on V, which implies that $\overline{\Gamma}^0$ acts irreducibly on V. Let $\tau_1, \overline{\tau}_1, \tau_2, \overline{\tau}_2$ be the embeddings of $\mathbb{Q}(\zeta_5)$ into $\overline{\mathbb{Q}}$, where

$$\tau_1|_{\mathbb{Q}(\sqrt{5})} \neq \tau_2|_{\mathbb{Q}(\sqrt{5})},$$

where $\overline{\tau}_j := \tau_j \circ \gamma$ for the complex conjugation γ of $\mathbb{Q}(\zeta_5)$, and let $V_\tau := V \otimes_{\mathbb{Q}(\zeta_5),\tau} \overline{\mathbb{Q}}$, so

$$V \otimes_{\mathbb{Q}} \overline{\mathbb{Q}} = V_{\tau_1} \oplus V_{\overline{\tau}_1} \oplus V_{\tau_2} \oplus V_{\overline{\tau}_2} \quad \text{and} \quad \dim V_\tau = 5.$$

Then

$$G_{\overline{\mathbb{Q}}} \cong SL(V_{\tau_1}) \times SL(V_{\tau_2}),$$

where $SL(V_{\tau_j})$ acts trivially on $V_{\tau_{3-j}}$ and $V_{\overline{\tau}_{3-j}}$, and acts on $V_{\overline{\tau}_j}$ as the dual of V_{τ_j} via the perfect pairing $V_{\tau_j} \times V_{\overline{\tau}_j} \to \overline{\mathbb{Q}}$ induced by $\langle \cdot, \cdot \rangle$. Since the standard representation of SL_5 is irreducible and not isomorphic to the dual representation, the four summands are pairwise non-isomorphic irreducible representations of $G_{\overline{\mathbb{Q}}}$. Let $W \subset V$ be a G-invariant $\mathbb{Q}$-subspace. Then $W \otimes \overline{\mathbb{Q}}$ is a sum of some of the four summands, and it is stable under $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, which permutes the four summands transitively since it acts transitively on $\{\tau_1, \overline{\tau}_1, \tau_2, \overline{\tau}_2\}$. Thus $W = 0$ or $W = V$.

Now let $u' \in U'$. Since g is dominant and generically finite and U' is irreducible, there exists a dense open subset $U_1 \subset U$ such that $g^{-1}(U_1) \to U_1$ is finite and étale. Since U' is connected, parallel transport along a path from u' to a point of $g^{-1}(U_1)$ conjugates the monodromy groups of $g^*\mathbb{V}$ and their Zariski closures, so we may assume that $u' \in g^{-1}(U_1)$. Let $u := g(u')$, and let Γ be the monodromy group of $\mathbb{V}$ at u. The image of $\pi_1(g^{-1}(U_1), u')$ in $\pi_1(U_1, u)$ has finite index, and the maps

$$\pi_1(U_1, u) \to \pi_1(U, u) \quad \text{and} \quad \pi_1(g^{-1}(U_1), u') \to \pi_1(U', u')$$

are surjective [Kol95, Proposition 2.10]. Thus the monodromy group of $g^*\mathbb{V}$ at u' is a finite index subgroup of Γ, so the connected algebraic monodromy group of $g^*\mathbb{V}$ at u' is $\overline{\Gamma}^0$, which acts irreducibly on $V = \mathbb{V}_{g(u')}$. □

5. A nef Hodge line bundle that is not semi-ample

Proposition 5.1. *Notation as in Construction-Proposition 4.1. Let $g : U' \to U$ be a dominant generically finite morphism from a smooth variety U', and let Z be a smooth projective variety containing U' as an open subset. Assume that $Z \backslash U'$ is an snc divisor and $g^*\mathbb{V}$ has unipotent local monodromy around $Z \backslash U'$. Let $j : U' \to Z$ be the inclusion, let $\overline{\mathcal{V}}$ be the Deligne canonical extension of $g^*\mathbb{V} \otimes \mathcal{O}_{U'}$ [Del70, Chapter II, Proposition 5.2], and let $\mathcal{A}_Z := j_* g^* \mathcal{A} \cap \overline{\mathcal{V}}$ be the Schmid extension of $g^*\mathcal{A}$. Then $\mathcal{A}_Z$ is nef, not big, and not semi-ample.*

Proof of Proposition 5.1. We divide the proof into two steps. Let ∇ be the Gauss–Manin connection of $\mathbb{V} \otimes \mathcal{O}_U$. We also write ∇ for the Gauss–Manin connection of any pullback of $\mathbb{V}$. Since ∇ preserves $g^*\mathbb{V}_\chi \otimes \mathcal{O}_{U'}$, we may let

$$\psi : T_{U'} \to \mathrm{Hom}\big(g^*\mathcal{A}, (g^*\mathbb{V}_\chi \otimes \mathcal{O}_{U'})/g^*\mathcal{A}\big)$$

be the $\mathcal{O}_{U'}$-linear map sending a local vector field v to the map $s \to \nabla_v s \bmod g^*\mathcal{A}$. For $x \in U'$, let ψ_x be the fiber of ψ at x.

Step 1. In this step we prove that $\mathcal{A}_Z$ is nef and not big.

Since $\chi^5 = 1$, we have $\mathbb{V}_{\chi^{-1}} = \mathbb{V}_{\chi^4}$. For any $u \in U$ and $i \in \{1, 2, 3, 4\}$, by Construction-Proposition 4.1(3),

$$\mathbb{V}_{\chi^i, u} = \mathbb{V}^{1,0}_{\chi^i, u} \oplus \mathbb{V}^{0,1}_{\chi^i, u} \quad \text{and} \quad \overline{\mathbb{V}^{p,q}_{\chi^i, u}} = \mathbb{V}^{q,p}_{\chi^{-i}, u} \text{ for any } (p, q) \in \{(1, 0), (0, 1)\}.$$

We set $\mathbb{V}^{p,q}_{\chi^i,u} := 0$ for any other $(p,q) \in \mathbb{Z}^2$ with $p+q=1$. For any $(p,q) \in \mathbb{Z}^2$ with $p+q=1$, let

$$\mathbb{V}'^{p,q}_{\chi,u} := \mathbb{V}^{p-2,q+2}_{\chi,u}, \quad \mathbb{V}'^{p,q}_{\chi^{-1},u} := \mathbb{V}^{p+2,q-2}_{\chi^{-1},u}, \quad \text{and} \quad \mathbb{V}'^{p,q}_{\chi^i,u} := \mathbb{V}^{p,q}_{\chi^i,u} \text{ for } i \in \{2,3\}$$

(compare [BFMT25, Remark 6.26]), and let

$$\mathbb{V}'^{p,q}_u := \bigoplus_{i=1}^{4} \mathbb{V}'^{p,q}_{\chi^i,u}.$$

Then

$$\mathbb{V}_u \otimes \mathbb{C} = \bigoplus_{p\in\mathbb{Z}} \mathbb{V}'^{p,1-p}_u \quad \text{and} \quad \overline{\mathbb{V}'^{p,q}_u} = \mathbb{V}'^{q,p}_u.$$

For $k \in \mathbb{Z}$, let $F^k\mathbb{V}_{\chi^i}$ be $\mathbb{V}_{\chi^i} \otimes \mathcal{O}_U$ if $k \le 0$, $\mathbb{V}^{1,0}_{\chi^i}$ if $k=1$, and 0 if $k \ge 2$. For $p \in \mathbb{Z}$, let

$$F^p\mathbb{V}' := F^{p-2}\mathbb{V}_{\chi} \oplus F^p\mathbb{V}_{\chi^2} \oplus F^p\mathbb{V}_{\chi^3} \oplus F^{p+2}\mathbb{V}_{\chi^{-1}},$$

a holomorphic subbundle of $\mathbb{V} \otimes \mathcal{O}_U$ with fiber $\bigoplus_{k\ge p} \mathbb{V}'^{k,1-k}_u$ at $u \in U$. Then

$$F^4\mathbb{V}' = 0, \qquad F^3\mathbb{V}' = \mathbb{V}^{1,0}_{\chi} = \mathcal{A}, \qquad \text{and} \qquad F^2\mathbb{V}' = \mathbb{V}_{\chi} \otimes \mathcal{O}_U.$$

Since ∇ preserves $\mathbb{V}_{\chi^i} \otimes \mathcal{O}_U$ and

$$\nabla F^k\mathbb{V}_{\chi^i} \subset F^{k-1}\mathbb{V}_{\chi^i} \otimes \Omega^1_U$$

for any i and k, we have

$$\nabla F^p\mathbb{V}' \subset F^{p-1}\mathbb{V}' \otimes \Omega^1_U$$

for any p.

Let Q be the polarization of $\mathbb{V}$ given by the cup product on $H^1(C_u, \mathbb{Q})$. Since μ_5 acts on C_u by automorphisms, Q is μ_5-invariant. Thus $Q(\mathbb{V}_{\chi^i,u}, \mathbb{V}_{\chi^j,u}) = 0$ unless $i+j=5$, and the subspaces $\mathbb{V}_{\chi^i,u}$, $i \in \{1,2,3,4\}$, are pairwise orthogonal for the sesquilinear form $(v,w) \to Q(v,\overline{w})$. For any i and (p,q), $\mathbb{V}'^{p,q}_{\chi^i,u} = \mathbb{V}^{p',q'}_{\chi^i,u}$ with $(p'-q')-(p-q) \in \{-4,0,4\}$, so $(\sqrt{-1})^{p-q} = (\sqrt{-1})^{p'-q'}$. Since Q satisfies the Hodge–Riemann bilinear relations for the decomposition $\bigoplus_{p,q} \mathbb{V}^{p,q}_{\chi^i,u}$, it satisfies them for the decomposition $\bigoplus_{p,q} \mathbb{V}'^{p,q}_u$. Therefore

$$\mathbb{V}' := (\mathbb{V} \otimes \mathbb{R}, F^\bullet\mathbb{V}', Q)$$

is a polarized real variation of Hodge structure of weight 1 on U.

The pullback $g^*\mathbb{V}'$ is a polarized real variation of Hodge structure on U' with

$$F^3g^*\mathbb{V}' = g^*\mathcal{A} \quad \text{and} \quad F^2g^*\mathbb{V}' = g^*\mathbb{V}_{\chi} \otimes \mathcal{O}_{U'},$$

so ψ is the map $T_{U'} \to \operatorname{Hom}(F^3, F^2/F^3)$ of [BBT23, Lemma 6.17] for $g^*\mathbb{V}'$. Since the underlying local system of $g^*\mathbb{V}'$ is $g^*\mathbb{V} \otimes \mathbb{R}$, $g^*\mathbb{V}'$ has unipotent local monodromy around $Z\backslash U'$, the Deligne canonical extension of $g^*\mathbb{V}'$ on Z is $\overline{\mathcal{V}}$, and the Schmid extension of $F^3g^*\mathbb{V}'$ is $\mathcal{A}_Z$. Since g is dominant and generically finite,

$$\operatorname{rank} T_{U'} = \dim U = 5 > 4 = \operatorname{rank} \operatorname{Hom}\big(g^*\mathcal{A}, (g^*\mathbb{V}_{\chi} \otimes \mathcal{O}_{U'})/g^*\mathcal{A}\big),$$

so ψ is not injective. By [BBT23, Lemma 6.17], $\det(\mathcal{A}_Z)$ is nef, and it is big if and only if ψ is injective as a map of $\mathcal{O}_{U'}$-modules. Thus $\mathcal{A}_Z = \det(\mathcal{A}_Z)$ is nef and not big.

Step 2. In this step we prove that $\mathcal{A}_Z$ is not semi-ample.

Suppose that $\mathcal{A}_Z$ is semi-ample. Let m be a positive integer such that $|\mathcal{A}_Z^{\otimes m}|$ is base point free, and let $Z \xrightarrow{f} Y \to \mathbb{P}^N$ be the Stein factorization of the morphism defined by $|\mathcal{A}_Z^{\otimes m}|$. Then f is a contraction onto a normal projective variety Y, and $\mathcal{A}_Z^{\otimes m} \cong f^*\mathcal{H}$ for an ample line bundle $\mathcal{H}$ on Y. Since $\mathcal{A}_Z$ is not big, $\dim Y < \dim Z = 5$. Let $y \in Y$ be a general point, let $Z_y := f^{-1}(y)$,

and let $Z_y^\circ := Z_y \cap U'$. Then Z_y is a smooth connected projective variety, Z_y° is a non-empty open subset of Z_y, and

$$\dim Z_y = 5 - \dim Y \geq 1, \qquad \mathcal{A}_Z^{\otimes m}|_{Z_y} \cong f^*\mathcal{H}|_{Z_y} \cong \mathcal{O}_{Z_y}.$$

We show that

$$\psi_x(v) = 0 \quad \text{for any } x \in Z_y^\circ \text{ and any } v \in T_x Z_y. \tag{5.1}$$

We may assume that $v \neq 0$. Let $n := \dim Z_y$, let $\mathcal{L}$ be a very ample line bundle on Z_y, and let $H_1, \dots, H_{n-1}$ be general members of $|\mathcal{L}|$ with $x \in H_j$ and $v \in T_x H_j$ for any j. Since $\mathcal{L}$ is very ample, $H_1 \cap \cdots \cap H_{n-1}$ is smooth of dimension 1 at x with tangent line $\mathbb{C}v$. Let Λ be the irreducible component of $H_1 \cap \cdots \cap H_{n-1}$ containing x (so $\Lambda = Z_y$ if $n = 1$), let $\nu : \widetilde{\Lambda} \to Z$ be the composition of the normalization of Λ with the inclusion $\Lambda \subset Z$, and let $\widetilde{\Lambda}^\circ := \nu^{-1}(U')$. Since $x \in \Lambda \cap U'$, $\widetilde{\Lambda} \backslash \widetilde{\Lambda}^\circ$ is a finite set. For any $t \in \widetilde{\Lambda} \backslash \widetilde{\Lambda}^\circ$, ν maps a small loop around t to a product of powers of small loops around the components of $Z \backslash U'$ through $\nu(t)$. The local monodromies of $g^*\mathbb{V}$ around the components of $Z \backslash U'$ through $\nu(t)$ commute and are unipotent, so $\nu^* g^* \mathbb{V}$ has unipotent local monodromy around t. By [BFMT25, Lemma 2.16(3)], the Schmid extension of $F^3 \nu^* g^* \mathbb{V}' = \nu^* g^* \mathcal{A}$ on $\widetilde{\Lambda}$ is $\nu^* \mathcal{A}_Z$. Since $\nu(\widetilde{\Lambda}) \subset Z_y$,

$$m \deg \nu^* \mathcal{A}_Z = \deg \nu^* \big(\mathcal{A}_Z^{\otimes m}|_{Z_y} \big) = 0,$$

so $\nu^* \mathcal{A}_Z$ is not big. Let

$$\psi_{\widetilde{\Lambda}} : T_{\widetilde{\Lambda}^\circ} \to \operatorname{Hom}\big(\nu^* g^* \mathcal{A}, (\nu^* g^* \mathbb{V}_\chi \otimes \mathcal{O}_{\widetilde{\Lambda}^\circ}) / \nu^* g^* \mathcal{A} \big)$$

be the $\mathcal{O}_{\widetilde{\Lambda}^\circ}$-linear map sending a local vector field ∂ to the map $s \to \nabla_\partial s \bmod \nu^* g^* \mathcal{A}$. Then $\psi_{\widetilde{\Lambda}} = \nu^* \psi \circ d\nu$. By [BBT23, Lemma 6.17], $\psi_{\widetilde{\Lambda}}$ is not injective. Since $T_{\widetilde{\Lambda}^\circ}$ is a line bundle on the integral curve $\widetilde{\Lambda}^\circ$, $\psi_{\widetilde{\Lambda}} = 0$. Since Λ is smooth at x, $\nu^{-1}(x)$ is a single point $\widetilde{x}$ and $d\nu(T_{\widetilde{x}} \widetilde{\Lambda}) = T_x \Lambda = \mathbb{C}v$. Thus $\psi_x(v) = 0$, which proves (5.1).

By (5.1), $\nabla_v s$ is a section of $g^* \mathcal{A}$ for any local section s of $g^* \mathcal{A}|_{Z_y^\circ}$ and any local vector field v on Z_y°. Thus the flat connection of $g^* \mathbb{V} \otimes \mathcal{O}_{Z_y^\circ}$ preserves the line subbundle $g^* \mathcal{A}|_{Z_y^\circ}$, and the flat sections of $g^* \mathcal{A}|_{Z_y^\circ}$ form a rank 1 local subsystem $\mathbb{W} \subset g^* \mathbb{V}_\chi|_{Z_y^\circ}$ with $\mathbb{W} \otimes \mathcal{O}_{Z_y^\circ} = g^* \mathcal{A}|_{Z_y^\circ}$. Since Z_y is smooth and connected, Z_y° is connected. Let $e_y : \mathcal{C}_y \to Z_y^\circ$ be the base change of the smooth projective morphism $e \circ c$ along $g|_{Z_y^\circ}$. Since the composition of the pullback

$$R^1 e_* \mathbb{Q}_{\mathcal{E}} \to R^1 (e \circ c)_* \mathbb{Q}_{\mathcal{C}}$$

with the trace map is multiplication by $\deg c = 5$,

$$R^1 (e \circ c)_* \mathbb{Q}_{\mathcal{C}} \cong R^1 e_* \mathbb{Q}_{\mathcal{E}} \oplus \mathbb{V}.$$

Thus $g^* \mathbb{V}|_{Z_y^\circ}$ is a direct summand of $R^1 e_{y*} \mathbb{Q}_{\mathcal{C}_y}$ that is pointwise a sub-Hodge structure (Construction-Proposition 4.1(3)), and $\mathbb{W}$ is a rank 1 complex local subsystem of $R^1 e_{y*} \mathbb{C}_{\mathcal{C}_y}$. By [Del71, Corollaire (4.2.8)(iii)(b)], a positive tensor power of $\mathbb{W}$ is a trivial local system. Fix a point $x \in Z_y^\circ$. Then the kernel of the monodromy representation $\pi_1(Z_y^\circ, x) \to GL(\mathbb{W}_x)$ is a subgroup of finite index, and it fixes a non-zero vector $w \in \mathbb{W}_x \subset \mathbb{V}_{g(x)} \otimes \mathbb{C}$.

By [Ver76, Corollaire 5.1], there exists a dense open subset $Y^\circ \subset Y$ over which $f|_{U'} : U' \to Y$ is a topologically locally trivial fibration. Since y is general, $y \in Y^\circ$. Let $U'^\circ := U' \cap f^{-1}(Y^\circ)$. Then $U'^\circ \to Y^\circ$ is a topologically locally trivial fibration with fiber Z_y° over y, and by the homotopy exact sequence, the image of $\pi_1(Z_y^\circ, x)$ in $\pi_1(U'^\circ, x)$ is the kernel of $\pi_1(U'^\circ, x) \to \pi_1(Y^\circ, y)$, hence a normal subgroup. Let $\rho : \pi_1(U', x) \to GL(\mathbb{V}_{g(x)})$ be the monodromy representation of $g^* \mathbb{V}$, let $\Pi := \rho(\pi_1(U', x))$, and let $\Pi_y \subset \Pi$ be the image of $\pi_1(Z_y^\circ, x)$. Since U' is smooth and $U'^\circ \subset U'$ is a dense open subset, $\pi_1(U'^\circ, x) \to \pi_1(U', x)$ is surjective [Kol95, Proposition 2.10]. Thus Π is the image of $\pi_1(U'^\circ, x)$, so Π_y is a normal subgroup of Π.

Let $\overline{\Pi}$ and $\overline{\Pi_y}$ be the Zariski closures of Π and Π_y in $GL(\mathbb{V}_{g(x)})$, and let $\overline{\Pi}^0$ and $\overline{\Pi_y}^0$ be their identity components. Since Π_y is normal in Π, $\overline{\Pi_y}^0$ is normal in $\overline{\Pi}$, and in particular $\overline{\Pi}^0$ normalizes $\overline{\Pi_y}^0$.

Since the image in Π_y of the kernel of $\pi_1(Z_y^\circ, x) \to GL(\mathbb{W}_x)$ has finite index and fixes w, its Zariski closure contains $\overline{\Pi_y}^0$, so $\overline{\Pi_y}^0$ fixes w. Let $V_0 \subset \mathbb{V}_{g(x)}$ be the $\mathbb{Q}$-subspace of vectors fixed by $\overline{\Pi_y}^0$. Since $\overline{\Pi_y}^0$ is defined over $\mathbb{Q}$, the subspace of $\mathbb{V}_{g(x)} \otimes \mathbb{C}$ fixed by $\overline{\Pi_y}^0$ is $V_0 \otimes \mathbb{C}$, which contains w. Thus $V_0 \neq 0$. Since $\overline{\Pi}^0$ normalizes $\overline{\Pi_y}^0$, $\overline{\Pi}^0$ preserves V_0. By Proposition 4.2(2), $V_0 = \mathbb{V}_{g(x)}$. Thus $\overline{\Pi_y}^0$ is trivial, and Π_y is finite.

For any $x' \in U'$, let

$$\overline{\nabla}_{x'} : T_{U',x'} \to \operatorname{Hom}(\mathbb{V}^{1,0}_{g(x')}, \mathbb{V}^{0,1}_{g(x')})$$

be the map of Proposition 4.2(1). The monodromy group of $g^*\mathbb{V}|_{Z_y^\circ}$ is Π_y. Since Π_y is finite, by [Del71, (4.1.3.3)] the Hodge structure of $g^*\mathbb{V}|_{Z_y^\circ}$ is locally constant, that is, $\nabla_v s$ is a section of $g^*\mathbb{V}^{1,0}$ for any local section s of $g^*\mathbb{V}^{1,0}|_{Z_y^\circ}$ and any local vector field v on Z_y°. Thus $\overline{\nabla}_{x'}$ vanishes on $T_{x'}Z_y$ for any $x' \in Z_y^\circ$. By Proposition 4.2(1), there exists a dense open subset $U_1' \subset U'$ such that $\overline{\nabla}_{x'}$ is injective for any $x' \in U_1'$. Since $f(U_1')$ contains a dense open subset of Y and y is general, there exists $x' \in Z_y \cap U_1' \subset Z_y^\circ$. Since $\dim Z_y \geq 1$, $T_{x'}Z_y \neq 0$, which contradicts the injectivity at x'. Thus $\mathcal{A}_Z$ is not semi-ample. □

6. Proof of Theorem 1.3

Proof of Theorem 1.3. Let U, $e : \mathcal{E} \to U$, P, D, $c : \mathcal{C} \to \mathcal{E}$, χ, $\mathbb{V}$, $\mathcal{A}$, and ξ be as in Construction-Proposition 4.1, and let $\phi := \xi^{-1}$. Since E_u is a curve of genus 1 for any $u \in U$, possibly shrinking U, we may assume that $\omega_{\mathcal{E}} \cong \mathcal{O}_{\mathcal{E}}$, and we take $K_{\mathcal{E}} := 0$. By Construction-Proposition 4.1(5), $\operatorname{div}(\phi) = 5P - D$, so

$$5\Big(K_{\mathcal{E}} + \frac{1}{5}D - P\Big) + \operatorname{div}(\phi) = 0.$$

Since $D \neq 0$ is reduced, $k(\frac{1}{5}D - P)$ is integral only if $5 \mid k$. Thus the integer r of Lemma 3.4 for $(\mathcal{E}, \frac{1}{5}D - P)$ is 5.

By Lemma 3.4 for $\pi_U := e$, $B_{X_U} := \frac{1}{5}D - P$, and ϕ, there exist a dense open subset $U_0 \subset U$, a finite étale morphism $g : U' \to U_0$ from a smooth variety U', and a smooth projective variety Z containing U' as an open subset with the property in Lemma 3.4. Fix an embedding $\mathcal{E} \times_U U' \subset \mathbb{P}^N \times U'$ over U'. A resolution of the closure of $\mathcal{E} \times_U U'$ in $\mathbb{P}^N \times Z$ that is an isomorphism over U' is a smooth projective variety X with a contraction $\pi : X \to Z$ such that $\pi^{-1}(U') = \mathcal{E} \times_U U'$. We let

- D_X and P_X be the closures in X of the pullbacks of D and P, and $B^h := \frac{1}{5}D_X - P_X$,
- ϕ_X be the pullback of ϕ to X, and
- K_X be a canonical divisor of X with $K_X|_{\pi^{-1}(U')} = 0$.

Then $(5(K_X + B^h) + \operatorname{div}(\phi_X))|_{\pi^{-1}(U')} = 0$, so no component of $5(K_X + B^h) + \operatorname{div}(\phi_X)$ dominates Z, and $B^v := -\frac{1}{5}(5(K_X + B^h) + \operatorname{div}(\phi_X))$ satisfies $B^v|_{\pi^{-1}(U')} = 0$.

Let X_η be the generic fiber of π, and $D_\eta := D_X|_{X_\eta}$ and $P_\eta := P_X|_{X_\eta}$. Then the following hold.

(i) X_η is a smooth curve of genus 1 and D_η is reduced.
(ii) P_η is a point of degree 1 on X_η, since P_X is the closure of a section of π over U'.
(iii) $P_\eta \notin \operatorname{Supp} D_\eta$, since $D \cap P = \emptyset$.
(iv) $D_\eta - 5P_\eta = -\operatorname{div}(\phi_X)|_{X_\eta}$, so $D_\eta \sim 5P_\eta$.

For any exceptional prime divisor T over X, $\dim \operatorname{center}_X T \le \dim X - 2 = \dim Z - 1$, so any prime divisor T over X whose center dominates Z is a horizontal prime divisor on X, and

$$a(T, X, B^h) = \begin{cases} 2 & \text{if } T = P_X, \\ \frac{4}{5} & \text{if } T \subset D_X, \\ 1 & \text{otherwise.} \end{cases}$$

Over U', $\operatorname{Supp} B^h$ is snc with coefficients $\frac{1}{5}$ and -1, so (X, B^h) is sub-klt over U'. Moreover, $\lceil \mathbf{A}^*(X, B^h) \rceil$ restricts to P_η on X_η by the display, so

$$\operatorname{rank} \pi_* \mathcal{O}_X(\lceil \mathbf{A}^*(X, B^h) \rceil) = h^0(X_\eta, \mathcal{O}_{X_\eta}(P_\eta)) = 1.$$

By Lemma 3.5, $\pi : (X, B) \to Z$ is an lc-trivial fibration for the $\mathbb{Q}$-divisor $B := B^h + B^v$, and $(\pi^{-1}(U'), B|_{\pi^{-1}(U')})$ is the base change of $(\mathcal{E}, \frac{1}{5}D - P)$ by g. Since $B|_{X_\eta} = \frac{1}{5}D_\eta - P_\eta$ and $\dim Z = \dim U' = \dim U = 5$, (1) and (2) hold.

By the property of Z in Lemma 3.4, the conditions of Set-up 3.1 hold for π, $\Sigma := Z \backslash U'$, ϕ_X, the pullback of $K_{\mathcal{E}}$, and a μ_5-equivariant resolution $X'' \to X'$, where X' is the normalization of X° in $\mathbb{C}(X)(\phi_X^{1/5})$ as in Set-up 3.1(4), with $d = 1$, $r = 5$, $X^\circ = \mathcal{E} \times_U U'$, and $(B^\circ)^{=1} = 0$, so $E'' = \emptyset$. Since $\phi^{1/5} = (\xi^{1/5})^{-1}$ and $e \circ c$ is smooth, Construction-Proposition 4.1(5) implies $X' = \mathcal{C} \times_U U'$. For a general point $u' \in U'$, the resolution $X'' \to X'$ restricts to a birational morphism $X''_{u'} \to C_{g(u')}$ between smooth projective curves, which is an isomorphism. Thus the μ_5-equivariant morphism of local systems

$$g^* R^1 (e \circ c)_* \mathbb{C}_{\mathcal{C}} \to \mathbb{H}$$

induced by $X'' \to X'$ is an isomorphism on U'. By Construction-Proposition 4.1(5), $\zeta \cdot \phi_X^{1/5} = \chi(\zeta) \phi_X^{1/5}$ for any $\zeta \in \mu_5$, so the character χ of Set-up 3.1 is the character χ of Construction-Proposition 4.1. Since χ is non-trivial and the μ_5-invariant part of $H^1(C_u, \mathbb{C})$ is the pullback of $H^1(E_u, \mathbb{C})$, the χ-isotypic part of $g^* R^1 (e \circ c)_* \mathbb{C}_{\mathcal{C}}$ is $g^* \mathbb{V}_\chi$. Thus, by the isomorphism $g^* R^1 (e \circ c)_* \mathbb{C}_{\mathcal{C}} \cong \mathbb{H}$ on U' and Construction-Proposition 4.1(4),

$$\mathbb{H}_\chi = g^* \mathbb{V}_\chi \quad \text{and} \quad F^1 \mathbb{H}_\chi = g^* \mathcal{A}.$$

Thus $(F^1 \mathbb{H}_\chi)_Z = \mathcal{A}_Z$, the Schmid extension of $g^* \mathcal{A}$, and by Proposition 3.3, $\mathbf{M}$ descends to Z and

$$\mathcal{O}_Z(m M_Z) \cong \mathcal{A}_Z^{\otimes m}$$

for some positive integer m. By Set-up 3.1(5), $\mathbb{H}$ has unipotent local monodromy around $Z \backslash U'$, and $g^* \mathbb{V} \otimes \mathbb{C} \subset \mathbb{H}$, so $g^* \mathbb{V}$ has unipotent local monodromy around $Z \backslash U'$. By Proposition 5.1 for the composite $U' \to U_0 \subset U$, $\mathcal{A}_Z$ is nef and not semi-ample. Thus M_Z is nef and not semi-ample, which is (3).

Since $\mathbf{M}$ descends to Z and M_Z is not semi-ample, $\mathbf{M}_{Z'} = f^* M_Z$ is not semi-ample for any projective birational morphism $f : Z' \to Z$ from a normal variety, so $\mathbf{M}$ is not $\boldsymbol{b}$-semi-ample. □

7. Realizing the Hodge line bundle by sub-pairs on the weighted projective plane

The generic fiber of the fibration in Theorem 1.3 is a curve of genus 1, while Corollary 1.4 requires the generic fiber to be of Fano type (Definition 8.2).

Construction-Proposition 7.1. Notation as in Construction-Proposition 4.1. Let $S := \mathbb{P}(1, 2, 3)$ with weighted coordinates x, y, z of weights $1, 2, 3$, and $H := (x = 0)$. Then there exist a dominant generically finite morphism $g : U' \to U$ from a smooth variety U', a smooth projective variety Z containing U' as an open subset such that $Z \backslash U'$ is an snc divisor and $g^* \mathbb{V}$ has unipotent local monodromy around $Z \backslash U'$, and an lc-trivial fibration $\pi : (S \times Z, B) \to Z$ with B a $\mathbb{Q}$-divisor and π the second projection, satisfying the following.

(1) On the generic fiber of π, $B = E_S + \frac{1}{5} D_S - H$, where $E_S \in |6H|$ is a smooth curve and $D_S \in |5H|$ does not contain H.

(2) The moduli $\boldsymbol{b}$-divisor $\mathbf{M}$ of π descends to Z, and $\mathcal{O}_Z(mM_Z) \cong \mathcal{A}_Z^{\otimes m}$ for some positive integer m such that mM_Z is integral, where $\mathcal{A}_Z$ is the Schmid extension of $g^*\mathcal{A}$.

Proof. We divide the proof into five steps. Let $v_2 := [0:1:0]$, $v_3 := [0:0:1]$, $y_1 := y/x^2$, and $z_1 := z/x^3$. Thus $S\backslash H$ is the affine plane with coordinates y_1, z_1.

Step 1. The fibers of e are curves of genus 1 and P is a section of e, so the generic fiber of e has a Weierstrass equation with origin given by P. Thus there exist a dense open subset $U_1 \subset U$, regular functions a_4, a_6 on U_1, and an isomorphism ι over U_1 from $e^{-1}(U_1)$ onto

$$E_S := (\sigma_E = 0) \subset S \times U_1, \quad \text{where} \quad \sigma_E := z^2 - y^3 - a_4x^4y - a_6x^6,$$

such that $\iota(P \cap e^{-1}(U_1)) = \{[0:1:1]\} \times U_1$ and, for any $u \in U_1$, the restrictions of y_1 and z_1 to $\iota(E_u)$ have poles of orders 2 and 3 at $\iota(P_u)$. Setting $x = 0$ in σ_E, we obtain $E_S \cap (H \times U_1) = \{[0:1:1]\} \times U_1$. In particular, E_S is disjoint from $\{v_2, v_3\} \times U_1$. Near $[0:1:1]$, the functions x and z are local coordinates of S on the chart $y = 1$, and $y_1 = x^{-2}$ there. Since $y_1|_{\iota(E_u)}$ has a pole of order 2 at $\iota(P_u)$,

$$H|_{\iota(E_u)} = \iota(P_u) \quad \text{for any } u \in U_1.$$

The monomials of degree 5 in x, y, z are $x^5, x^3y, xy^2, x^2z, yz$. Their quotients by x^5 are $1, y_1, y_1^2, z_1, y_1z_1$, whose restrictions to $\iota(E_u)$ are regular outside $\iota(P_u)$ and have poles of orders $0, 2, 4, 3, 5$ at $\iota(P_u)$. Since $h^0(E_u, \mathcal{O}_{E_u}(5P_u)) = 5$, the five restrictions form a basis of $H^0(E_u, \mathcal{O}_{E_u}(5P_u))$ for any $u \in U_1$, so their pullbacks by ι form a basis of the locally free sheaf $e_*\mathcal{O}_{\mathcal{E}}(5P)|_{U_1}$. By Construction-Proposition 4.1(5), $\operatorname{div}(\xi) = D - 5P$, so ξ is a section of $\mathcal{O}_{\mathcal{E}}(5P)$ over $e^{-1}(U_1)$, and there exist regular functions $\lambda_1, \dots, \lambda_5$ on U_1 such that

$$\sigma_D := \lambda_1x^5 + \lambda_2x^3y + \lambda_3xy^2 + \lambda_4x^2z + \lambda_5yz \quad \text{satisfies} \quad \frac{\sigma_D}{x^5} \circ \iota = \xi \text{ on } e^{-1}(U_1).$$

We let $D_S := (\sigma_D = 0) \subset S \times U_1$. Since $\operatorname{div}(\xi) = D - 5P$ and $H|_{\iota(E_u)} = \iota(P_u)$, we have $D_S|_{E_S} = \iota(D \cap e^{-1}(U_1))$. Since $D_u \cap P_u = \emptyset$, $\xi|_{E_u}$ has a pole of order 5 at P_u. Among $1, y_1, y_1^2, z_1, y_1z_1$, only y_1z_1 has a pole of order 5 at $\iota(P_u)$, so λ_5 has no zero on U_1. Hence $\sigma_D|_{x=0} = \lambda_5yz$ and $D_S \cap (H \times U_1) = \{v_2, v_3\} \times U_1$. Moreover, on $(S\backslash H) \times U_1$,

$$\frac{\sigma_D}{x^5} = \lambda_5(y_1 - t_0)z_1 + \lambda_1 + \lambda_2y_1 + \lambda_3y_1^2, \quad \text{where} \quad t_0 := -\frac{\lambda_4}{\lambda_5}. \tag{7.1}$$

Step 2. Let $\mathbf{k}$ be an algebraic closure of $\mathbb{C}(U_1)$. In this step we work over $\mathbf{k}$, keep the notation S, H, E_S, D_S for the base changes to $\mathbf{k}$ of S, H and of the generic fibers of E_S, D_S over U_1, and let

$$B_S := E_S + \frac{1}{5}D_S - H.$$

By **Step 1**, E_S is a smooth curve contained in $S\backslash\{v_2, v_3\}$, $H|_{E_S} = [0:1:1]$, $D_S \cap H = \{v_2, v_3\}$, and $D_S|_{E_S}$ is reduced of degree 5, since D is étale over U. The open subset $(x \neq 0)$ of S is $\mathbb{A}^2$, and the open subsets $(y \neq 0)$ and $(z \neq 0)$ are the quotients of the affine plane with coordinates x, z by μ_2 acting by -1 and of the affine plane with coordinates x, y by μ_3 acting with weights $(1, 2)$. Hence $\operatorname{Sing} S = \{v_2, v_3\}$, and the singularities $v_2 \in S$ and $v_3 \in S$ are of types $\frac{1}{2}(1, 1)$ and $\frac{1}{3}(1, 2)$. By [Dol82, Theorem 3.3.4], $K_S \sim -6H$. Since $E_S \sim 6H$ and $D_S \sim 5H$,

$$K_S + B_S \sim_{\mathbb{Q}} (-6 + 6 + 1 - 1)H = 0.$$

We show that $\operatorname{Supp} B_S \cap S_{\mathrm{reg}}$ is an snc divisor, where $S_{\mathrm{reg}} := S \setminus \operatorname{Sing} S$. Since $S_{\mathrm{reg}} \cap H = H\backslash\{v_2, v_3\}$ is disjoint from D_S, we only need to consider $D_S \cap (S\backslash H)$. By (7.1), a singular point of $D_S \cap (S\backslash H)$ satisfies $y_1 = t_0$ and $\lambda_1 + \lambda_2t_0 + \lambda_3t_0^2 = 0$. Thus either $D_S \cap S_{\mathrm{reg}}$ is smooth, or $\lambda_1 + \lambda_2t_0 + \lambda_3t_0^2 = 0$ and

$$\frac{\sigma_D}{x^5} = (y_1 - t_0)(\lambda_5z_1 + \lambda_3y_1 + \lambda_2 + \lambda_3t_0),$$

in which case $D_S \cap S_{\mathrm{reg}}$ is the union of two smooth curves intersecting transversally at one point of $S \backslash H$. Since $D_S|_{E_S}$ is reduced, E_S intersects D_S transversally and $E_S \cap \operatorname{Sing} D_S = \emptyset$. Since $H|_{E_S} = [0:1:1]$ and $[0:1:1] \notin D_S$, E_S intersects H transversally at a point outside D_S. Hence $\operatorname{Supp} B_S \cap S_{\mathrm{reg}}$ is an snc divisor.

The singularities v_2, v_3 are Du Val. Let $q : \widetilde{S} \to S$ be the minimal resolution, Θ_1 the exceptional curve over v_2, and Θ_2, Θ_3 the exceptional curves over v_3. Then $\Theta_1^2 = \Theta_2^2 = \Theta_3^2 = -2$ and $\Theta_2 \cdot \Theta_3 = 1$. Near v_2 and v_3, q is the toric resolution in the coordinates x, z and x, y, and for $\Gamma_1 := (z = 0)$ and $\Gamma_2 = \Gamma_3 := (y = 0)$,

$$(\operatorname{mult}_{\Theta_1} q^*H, \operatorname{mult}_{\Theta_1} q^*\Gamma_1) = (\tfrac{1}{2}, \tfrac{1}{2}),$$
$$(\operatorname{mult}_{\Theta_2} q^*H, \operatorname{mult}_{\Theta_2} q^*\Gamma_2) = (\tfrac{1}{3}, \tfrac{2}{3}),$$
$$\text{and } (\operatorname{mult}_{\Theta_3} q^*H, \operatorname{mult}_{\Theta_3} q^*\Gamma_3) = (\tfrac{2}{3}, \tfrac{1}{3}).$$

Hence, for a curve $\Lambda \subset S$ and $i \in \{1, 2, 3\}$,

$$\operatorname{mult}_{\Theta_i} q^*\Lambda = \min\{a \operatorname{mult}_{\Theta_i} q^*H + b \operatorname{mult}_{\Theta_i} q^*\Gamma_i\},$$

where the minimum runs over the monomials $x^a z^b$ (for $i = 1$) or $x^a y^b$ (for $i \in \{2, 3\}$) of a local equation of the preimage of Λ in the affine plane near v_2 (for $i = 1$) or v_3 (for $i \in \{2, 3\}$). The preimages of H and D_S are defined by x and by

$$\lambda_3 x + \lambda_5 z + \lambda_4 x^2 z + \lambda_2 x^3 + \lambda_1 x^5 \text{ near } v_2, \qquad \lambda_5 y + \lambda_4 x^2 + \lambda_3 x y^2 + \lambda_2 x^3 y + \lambda_1 x^5 \text{ near } v_3.$$

Since $\lambda_5 \neq 0$, we obtain $(\operatorname{mult}_{\Theta_i} q^*D_S)_{i=1}^3 = (\frac{1}{2}, \frac{2}{3}, \frac{1}{3})$ and $(\operatorname{mult}_{\Theta_i} q^*H)_{i=1}^3 = (\frac{1}{2}, \frac{1}{3}, \frac{2}{3})$. Let $D' := q_*^{-1} D_S$ and $H' := q_*^{-1} H$, and denote $q^{-1}(E_S) \cong E_S$ again by E_S. Since q is crepant, $K_{\widetilde{S}} + \widetilde{B} = q^*(K_S + B_S)$ for

$$\widetilde{B} := q^*B_S = E_S + \frac{1}{5}D' - H' - \frac{2}{5}\Theta_1 - \frac{1}{5}\Theta_2 - \frac{3}{5}\Theta_3.$$

From $q^*D_S \cdot \Theta_i = q^*H \cdot \Theta_i = 0$ for any $i \in \{1, 2, 3\}$ we obtain

$$D' \cdot \Theta_1 = D' \cdot \Theta_2 = H' \cdot \Theta_1 = H' \cdot \Theta_3 = 1 \quad \text{and} \quad D' \cdot \Theta_3 = H' \cdot \Theta_2 = 0.$$

Near v_2, the preimages of D_S and H are smooth with the distinct tangent lines $(\lambda_3 x + \lambda_5 z = 0)$ and $(x = 0)$, and Θ_1 is the curve of tangent lines at the origin, so $D' \cap \Theta_1 \neq H' \cap \Theta_1$. Near v_3, $D' \cap \Theta_3 = H' \cap \Theta_2 = \emptyset$ since $D' \cdot \Theta_3 = H' \cdot \Theta_2 = 0$, and D' intersects Θ_2 and H' intersects Θ_3 transversally at one point since $D' \cdot \Theta_2 = H' \cdot \Theta_3 = 1$. Since moreover $D_S \cap H = \{v_2, v_3\}$ and $\operatorname{Supp} B_S \cap S_{\mathrm{reg}}$ is snc, $\operatorname{Supp} \widetilde{B}$ is an snc divisor.

Since $(\widetilde{S}, \widetilde{B})$ is log smooth with coefficients at most 1, it is sub-lc, so (S, B_S) is sub-lc. Hence $\lceil \mathbf{A}^*(S, B_S) \rceil \geq 0$ by Definition 2.3, so the constant function 1 is a section of $\mathcal{O}(\lceil \mathbf{A}^*(S, B_S) \rceil)$ on any higher model of S. Conversely, the trace of $\lceil \mathbf{A}^*(S, B_S) \rceil$ on S is H, because $a(E_S, S, B_S) = 0$, $a(H, S, B_S) = 2$, $a(\Lambda, S, B_S) = \frac{4}{5}$ for any component Λ of D_S, and $a(\Lambda, S, B_S) = 1$ for any other prime divisor Λ on S. So any section is a rational function f on S with $\operatorname{div}(f) + H \geq 0$, that is, $f \in x^{-1} H^0(S, \mathcal{O}_S(1)) = \mathbf{k}$. Hence

$$\dim_{\mathbf{k}} H^0(S, \mathcal{O}_S(\lceil \mathbf{A}^*(S, B_S) \rceil)) = 1.$$

Step 3. Let $B_{S \times U_1} := E_S + \frac{1}{5} D_S - H \times U_1$ and

$$\phi := \frac{x^{35}}{\sigma_E^5 \sigma_D}.$$

Since $\operatorname{div}(\sigma_E / x^6) = E_S - 6H \times U_1$ and $\operatorname{div}(\sigma_D / x^5) = D_S - 5H \times U_1$, we have

$$\operatorname{div}(\phi) = 35H \times U_1 - 5E_S - D_S,$$

that is,

$$5(-6H \times U_1 + B_{S \times U_1}) + \operatorname{div}(\phi) = 0. \tag{7.2}$$

Let $w := (x^6/\sigma_E)\phi^{-1/5}$, a rational function on the cyclic cover of $S \times U_1$ given by a fifth root of ϕ. Then $w^5 = \sigma_D/x^5$ and $\phi^{1/5} = x^6/(\sigma_E w)$. By (7.1),

$$z_1 = \frac{w^5 - \lambda_1 - \lambda_2 y_1 - \lambda_3 y_1^2}{\lambda_5(y_1 - t_0)},$$

so, over $y_1 \neq t_0$, the hypersurface

$$(w^5 = \sigma_D/x^5) \subset \mathbb{A}^1 \times (S \backslash H) \times U_1$$

is the graph of z_1 as a function of (y_1, w) and the point of U_1. Since

$$\dim\big((w^5 = \sigma_D/x^5) \cap (y_1 = t_0)\big) = \dim U_1 + 1,$$

no component of $(w^5 = \sigma_D/x^5)$ is contained in $(y_1 = t_0)$. Thus the hypersurface $(w^5 = \sigma_D/x^5)$ is irreducible and birational to $\mathbb{A}^2 \times U_1$ via (y_1, w) and the projection to U_1. Since $\phi^{1/5} = x^6/(\sigma_E w)$, the cyclic cover of $S \times U_1$ given by a fifth root of ϕ is birational to $(w^5 = \sigma_D/x^5)$ via w.

Step 4. In this step we construct g, Z, and B, prove (1), and show that $\mathbf{M}$ descends to Z with $\mathcal{O}_Z(mM_Z) \cong ((F^2\mathbb{H}_\chi)_Z)^{\otimes m}$.

Let $B_{\widetilde{S}\times U_1} := (q \times \mathrm{id}_{U_1})^* B_{S\times U_1}$. The base change of $B_{\widetilde{S}\times U_1}$ to the geometric generic fiber is $\widetilde{B}$. By **Step 2** and generic smoothness, there exists a dense open subset $U_2 \subset U_1$ such that, for

$$B_{\widetilde{S}\times U_2} := B_{\widetilde{S}\times U_1}|_{\widetilde{S}\times U_2},$$

$\operatorname{Supp} B_{\widetilde{S}\times U_2}$ is an snc divisor. Possibly shrinking U_2, we may assume that $\Omega^5_{U_2} \cong \mathcal{O}_{U_2}$. We fix a generator ω of $\Omega^5_{U_2}$, so that $K_{U_2} := 0$ is a canonical divisor of U_2. Let $K_{\widetilde{S}} := q^*(-6H)$, a canonical divisor of $\widetilde{S}$ since q is crepant, and $K_{\widetilde{S}\times U_2} := K_{\widetilde{S}} \times U_2$. By (7.2),

$$5(K_{\widetilde{S}\times U_2} + B_{\widetilde{S}\times U_2}) + \operatorname{div}(\phi) = 0,$$

where ϕ also denotes the pullback of ϕ. Let X_η be the generic fiber of $\widetilde{S} \times U_2 \to U_2$. Since D' has coefficient $\frac{1}{5}$, $k(K_{X_\eta} + B_{\widetilde{S}\times U_2}|_{X_\eta})$ is integral only if $5 \mid k$, so by (7.2),

$$\min\{k \in \mathbb{N}^+ \mid k(K_{X_\eta} + B_{\widetilde{S}\times U_2}|_{X_\eta}) \sim 0\} = 5.$$

By Lemma 3.4 for $\widetilde{S} \times U_2 \to U_2$, $B_{\widetilde{S}\times U_2}$, and ϕ, there exist a dense open subset $U_3 \subset U_2$, a finite étale morphism $g : U' \to U_3$ from a smooth variety U', and a smooth projective variety Z containing U' as an open subset, with the property in Lemma 3.4. We also regard g as a morphism to U. Then g is dominant and generically finite.

Since g is étale, $g^*\omega$ is a generator of $\Omega^5_{U'}$. We let K_Z be the divisor of $g^*\omega$ on Z, so that $K_Z|_{U'} = 0$, and let $K_{S\times Z} := -6H \times Z + S \times K_Z$. Let B^h be the closure in $S \times Z$ of the pullback of $B_{S\times U_2} := B_{S\times U_1}|_{S\times U_2}$ under $\mathrm{id}_S \times g$. Since S has only quotient singularities, $S \times Z$ is $\mathbb{Q}$-factorial. By (7.2) and $K_Z|_{U'} = 0$, the divisor $5(K_{S\times Z} + B^h) + \operatorname{div}(\phi)$ is zero on $S \times U'$, so no component of $5(K_{S\times Z} + B^h) + \operatorname{div}(\phi)$ dominates Z. Since U' is dense in Z and $(S \times U', B^h|_{S\times U'})$ is the base change of $(S \times U_2, B_{S\times U_2})$ by the finite morphism g (we omit the passage from $\mathbf{k}$, an algebraic closure of $\mathbb{C}(Z)$, to $\mathbb{C}(Z)$ by flat base change on the log resolution $\widetilde{S}$ of **Step 2**), **Step 2** shows that $(S \times Z, B^h)$ is sub-lc over the generic point of Z and

$$\operatorname{rank} \pi_* \mathcal{O}_{S\times Z}(\lceil \mathbf{A}^*(S \times Z, B^h)\rceil) = 1.$$

By Lemma 3.5, $\pi : (S \times Z, B) \to Z$ is an lc-trivial fibration with $K_{S\times Z} + B \sim_{\mathbb{Q}} 0$, where

$$B := B^h + B^v \quad \text{and} \quad B^v := -\frac{1}{5}\big(5(K_{S\times Z} + B^h) + \operatorname{div}(\phi)\big),$$

and $B^v|_{S\times U'} = 0$. On the generic fiber of π, $B = B^h = E_S + \frac{1}{5}D_S - H$, where $E_S \in |6H|$ is smooth and $D_S \in |5H|$ does not contain H by **Step 1**. Thus (1) holds.

Let $\widetilde{\pi} : \widetilde{S} \times Z \to Z$ be the second projection, $B_{\widetilde{S}\times Z} := (q \times \mathrm{id}_Z)^* B$, and

$$K_{\widetilde{S}\times Z} := K_{\widetilde{S}} \times Z + \widetilde{S} \times K_Z.$$

Since $q \times \mathrm{id}_Z$ is crepant, $\widetilde{\pi}$ is an lc-trivial fibration for the sub-pair $(\widetilde{S} \times Z, B_{\widetilde{S}\times Z})$, and the discriminant and moduli $\boldsymbol{b}$-divisors of $\widetilde{\pi}$ are those of π. Since $B^v|_{S\times U'} = 0$, the sub-pair

$$(\widetilde{\pi}^{-1}(U'), B_{\widetilde{S}\times Z}|_{\widetilde{\pi}^{-1}(U')})$$

is the base change of $(\widetilde{S} \times U_2, B_{\widetilde{S}\times U_2})$ by g. By Lemma 3.4, the conditions of Set-up 3.1 hold for $\widetilde{\pi}$, $\Sigma := Z\backslash U'$, the pullbacks of ϕ and $K_{\widetilde{S}\times U_2}$, and some μ_5-equivariant resolution h, with $d = 2$ and $r = 5$. In particular $Z\backslash U'$ is an snc divisor. By Proposition 3.3, $\mathbf{M}$ descends to Z and

$$\mathcal{O}_Z(mM_Z) \cong ((F^2\mathbb{H}_\chi)_Z)^{\otimes m}$$

for some positive integer m such that mM_Z is integral, where $\mathbb{H}_\chi$ is the local system of Set-up 3.1 for $\widetilde{\pi}$.

Step 5. In this step we show that $(F^2\mathbb{H}_\chi)_Z \cong \mathcal{A}_Z$ and that $g^*\mathbb{V}$ has unipotent local monodromy around $Z\backslash U'$.

We use the notation of Set-up 3.1 for $\widetilde{\pi}$: $Z^\circ = U'$, $X^\circ = \widetilde{S} \times U'$, $f : X' \to X^\circ$, $\pi'' : X'' \to U'$, and E''. We keep the notation E_S, D', H', ι, w for their base changes by g. Then $(B^\circ)^{=1} = E_S$. Near E_S, we have $5(K_{X^\circ} + B^\circ) = -35H' \times U' + 5E_S + D'$ and $D'|_{E_S} = \iota(D)$, and the divisor D' is smooth and intersects E_S transversally since $\iota(D)$ is étale over U'. Hence X' is smooth near $f^{-1}(E_S)$, f is flat there, and $f^{-1}(E_S) \to E_S$ is a cyclic cover of degree 5 branched along $\iota(D)$. Since $w^5 = \sigma_D/x^5$ and $(\sigma_D/x^5) \circ \iota = \xi$, Construction-Proposition 4.1(5) implies that ι lifts to an isomorphism ι' over U' from $\mathcal{C} \times_U U'$ onto $f^{-1}(E_S)$. We let μ_5 act on X' so that ι' is μ_5-equivariant. The new action differs from the action of Set-up 3.1(4) by an automorphism of μ_5, so h is still μ_5-equivariant and $\mathbb{H}_\chi$ does not change. Under ι', w corresponds to $\xi^{1/5}$. Since $\phi^{1/5} = x^6/(\sigma_E w)$ and x^6/σ_E is a rational function on X°, Construction-Proposition 4.1(5) shows that μ_5 acts on $\phi^{1/5}$ by χ, so the character χ of Set-up 3.1 is the character χ of Construction-Proposition 4.1. For $u \in U'$, ι' restricts to an isomorphism from $C_{g(u)}$ onto the fiber of $f^{-1}(E_S)$ over u.

Let $u \in U'$ be a general point. By **Step 3**, the rational map $X'' \dashrightarrow \mathbb{A}^2 \times U'$ given by (y_1, w) and π'' is birational, so X''_u is a smooth projective rational surface. Thus $H^1(X''_u) = H^3(X''_u) = 0$ and $H^2(X''_u)$ is of type $(1, 1)$. Since X'_u is smooth near $C_{g(u)}$, $h_u : X''_u \to X'_u$ is near $C_{g(u)}$ a composition of blow-ups of points. Hence $E''_u = h_u^{-1}(C_{g(u)})$ consists of $(h_u)_*^{-1}C_{g(u)} \cong C_{g(u)}$ and of smooth rational curves, and the dual graph of E''_u is a tree. Let $E''^{[k]}_u$ be the disjoint union of the k-fold intersections of components of E''_u. By the weight spectral sequence [Del71, Section 3.2], the weights of $H^2(X''_u\backslash E''_u)$ are in $\{2, 3, 4\}$ and

$$\begin{aligned} &\mathrm{gr}_2^W H^2(X''_u\backslash E''_u) = \mathrm{Im}\big(H^2(X''_u) \to H^2(X''_u\backslash E''_u)\big),\\ &\mathrm{gr}_3^W H^2(X''_u\backslash E''_u) \cong H^1(C_{g(u)})(-1),\\ \text{and}\quad &\mathrm{gr}_4^W H^2(X''_u\backslash E''_u) = 0. \end{aligned}$$

Here the isomorphism is the residue along $(h_u)_*^{-1}C_{g(u)}$, which is an isomorphism because H^1 of a rational curve and $H^3(X''_u)$ vanish, and $\mathrm{gr}_4^W = 0$ because the Gysin map $H^0(E''^{[2]}_u) \to H^2(E''^{[1]}_u)$ is injective for a tree. The residue and the Gysin maps are μ_5-equivariant. By Set-up 3.1(4), the strata of E'' are smooth over U', so W is a filtration of $\mathbb{H}$ by sub-local systems, the residue is a morphism of variations of mixed Hodge structure, and the three descriptions of gr_k^W above hold for the variations over U'. Since $H^1(C_{g(u)}) = H^1(E_{g(u)}) \oplus \mathbb{V}_{g(u)}$ with μ_5 acting trivially on $H^1(E_{g(u)})$, and χ is non-trivial, we obtain

$$\mathbb{H}_\chi = W_3\mathbb{H}_\chi, \qquad F^2\,\mathrm{gr}_2^W\,\mathbb{H}_\chi = 0, \qquad \text{and} \qquad \mathrm{gr}_3^W\,\mathbb{H}_\chi \cong g^*\mathbb{V}_\chi(-1).$$

Since $\mathbb{H}_\chi = W_3\mathbb{H}_\chi$, the filtration F on $\mathrm{gr}_3^W\,\mathbb{H}_\chi$ is the image of F on $\mathbb{H}_\chi$. Since moreover

$$F^2W_2\mathbb{H}_\chi = F^2\,\mathrm{gr}_2^W\,\mathbb{H}_\chi = 0,$$

the projection to $\mathrm{gr}_3^W \mathbb{H}_\chi$ restricts to an isomorphism

$$F^2\mathbb{H}_\chi \to F^2 \mathrm{gr}_3^W \mathbb{H}_\chi.$$

By the equality $(F^2\mathbb{H}_\chi)_Z = F^2(\mathcal{V}_Y^{\mathrm{tr}})_\chi$ in [BFMT25, Construction-Definition 6.23], where $\mathcal{V}_Y^{\mathrm{tr}}$ is the Deligne extension [BFMT25, Section 2.4] of the transcendental part V_Y^{tr} and the χ-isotypic part of V_Y^{tr} is a sub-variation of $\mathrm{gr}_3^W \mathbb{H}_\chi$ containing $F^2 \mathrm{gr}_3^W \mathbb{H}_\chi$ [BFMT25, Definition 2.9(3) and Remark 6.26] whose Deligne extension is a subbundle of that of $\mathrm{gr}_3^W \mathbb{H}_\chi$ [Del70, Chapter II, Proposition 5.2(d)], the Schmid extension $(F^2\mathbb{H}_\chi)_Z$ is the Schmid extension of $F^2 \mathrm{gr}_3^W \mathbb{H}_\chi$ in $\mathrm{gr}_3^W \mathbb{H}_\chi$. A Tate twist does not change the underlying line bundle, and

$$F^2 \mathrm{gr}_3^W \mathbb{H}_\chi \cong F^2(g^*\mathbb{V}_\chi(-1)) = g^*\mathbb{V}_\chi^{1,0} = g^*\mathcal{A}.$$

Thus

$$(F^2\mathbb{H}_\chi)_Z \cong \mathcal{A}_Z,$$

and by **Step 4**, $\mathcal{O}_Z(mM_Z) \cong \mathcal{A}_Z^{\otimes m}$, which is (2).

By Set-up 3.1(5), $R^2\pi''_*\mathbb{Z}_{X''\setminus E''}$ has unipotent local monodromy around $Z\setminus U'$. Hence the subquotient gr_3^W of $R^2\pi''_*\mathbb{Q}_{X''\setminus E''}$ has unipotent local monodromy around $Z\setminus U'$. The residue is an isomorphism

$$\mathrm{gr}_3^W R^2\pi''_*\mathbb{Q}_{X''\setminus E''} \to g^*R^1(e\circ c)_*\mathbb{Q}_{\mathcal{C}}(-1),$$

and $g^*\mathbb{V}(-1)$ is a sub-local system of the right-hand side, so $g^*\mathbb{V}(-1)$ has unipotent local monodromy around $Z\setminus U'$. Thus $g^*\mathbb{V}$ has unipotent local monodromy around $Z\setminus U'$. □

8. Proof of Corollary 1.4

Definition 8.1 (cf. [HJL22, Definitions 2.5 and 2.7])**.** Let $(X/Z \ni z, B)$ be a sub-pair with contraction $\pi : X \to Z$, and let $\overline{z}$ be the closure of z. The *minimal log discrepancy* of $(X/Z \ni z, B)$ is

$$\mathrm{mld}(X/Z \ni z, B) := \inf\{a(E, X, B) \mid E \text{ is a prime divisor over } X \text{ with } \pi(\mathrm{center}_X E) = \overline{z}\}.$$

For a real number $\delta \geq 0$, the sub-pair $(X/Z \ni z, B)$ is *δ-lc* if $\mathrm{mld}(X/Z \ni z, B) \geq \delta$, and a sub-pair (X, B) is *δ-lc* if $(X \ni x, B)$ is δ-lc for any (not necessarily closed) point $x \in X$ of codimension ≥ 1.

Definition 8.2 (cf. [HJL22, Definition 2.16])**.** A normal projective variety X over a field $k \supset \mathbb{C}$ is *of Fano type* if there exists a klt pair (X, B) over k (Definition 2.1 with $\mathbb{C}$ replaced by k) such that $-(K_X + B)$ is ample.

Proof of Corollary 1.4. We divide the proof into several steps.

Step 1. Let $g : U' \to U$ and $\pi_W : (S \times W, \Delta) \to W$ be the morphism and the lc-trivial fibration given by Construction-Proposition 7.1, where W and Δ are the smooth projective variety and the $\mathbb{Q}$-divisor denoted by Z and B there. Let B_W and M_W be the discriminant and the moduli part of π_W, and let L_W be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor as in Definition 2.5. Since $\dim U = 5$ and g is dominant and generically finite, $\dim W = 5$. By Construction-Proposition 7.1(2) and Proposition 5.1, $\mathbf{M}$ descends to W, and M_W is nef, not big, and not semi-ample.

Since M_W is not semi-ample, $\mathrm{Bs}|knM_W| \subset \mathrm{Bs}|nM_W|$ for any $k \in \mathbb{N}^+$, and W is quasi-compact, there exists a closed point $p \in W$ satisfying

$$p \in \mathrm{Bs}|nM_W| \quad \text{for any positive integer } n \text{ such that } nM_W \text{ is integral.} \tag{8.1}$$

Next we show that for any $\mathbb{R}$-divisor $D \geq 0$ with $D \sim_{\mathbb{R}} M_W$ we have $p \in \mathrm{Supp}\, D$. Write

$$D = \sum_{i=1}^{N} \gamma_i D_i \quad \text{and} \quad D - M_W = \sum_{j=1}^{N'} \lambda_j \,\mathrm{div}(\psi_j),$$

where $D_1, \dots, D_N$ are distinct prime divisors, $\gamma_i > 0$, $\lambda_j \in \mathbb{R}$, and ψ_j are rational functions on W. Comparing the coefficients along the finitely many prime divisors contained in

$$\operatorname{Supp} D \cup \operatorname{Supp} M_W \cup \bigcup_{j=1}^{N'} \operatorname{Supp} \operatorname{div}(\psi_j),$$

the vector $(\gamma_1, \dots, \gamma_N, \lambda_1, \dots, \lambda_{N'})$ is a real solution of a system of linear equations with rational coefficients. Since the rational solutions of such a system are dense in the real solutions, there exists a rational solution $(\gamma'_1, \dots, \gamma'_N, \lambda'_1, \dots, \lambda'_{N'})$ with $\gamma'_i > 0$ for any i. Let n_0 be a positive integer such that $n_0 M_W$, $n_0\gamma'_i$, and $n_0\lambda'_j$ are integral for any i and j. Then

$$\sum_{i=1}^{N} n_0\gamma'_i D_i = n_0 M_W + \operatorname{div}\Big(\prod_{j=1}^{N'} \psi_j^{n_0\lambda'_j}\Big) \in |n_0 M_W|,$$

so $p \in \bigcup_{i=1}^{N} D_i = \operatorname{Supp} D$ by (8.1).

Step 2. Let A be a very ample Cartier divisor on W. Since M_W is nef, there exists a positive integer m such that $K_W + 5mA + M_W$ is ample. By Bertini, there exist $R_1, \dots, R_5 \in |mA|$ containing p such that local equations of $R_1, \dots, R_5$ at p form a regular system of parameters of W at p and $R_1 + \cdots + R_5$ is snc. Since $R_1 + \cdots + R_5 \sim 5mA$, $K_W + R_1 + \cdots + R_5 + M_W$ is ample. Let r be a positive integer such that $r(K_W + R_1 + \cdots + R_5 + M_W)$ is integral and very ample, let F' be a general member of $|r(K_W + R_1 + \cdots + R_5 + M_W)|$, and let $F := \frac{1}{r}F'$. Then $p \notin \operatorname{Supp} F$, $R_1 + \cdots + R_5 + F$ is snc, and we let

$$B'_W := R_1 + \cdots + R_5 - F.$$

Since $F' \in |r(K_W + R_1 + \cdots + R_5 + M_W)|$, there exists a rational function φ on W with

$$r(K_W + B'_W + M_W) = \operatorname{div}_W(\varphi). \tag{8.2}$$

Since B'_W is snc with coefficients in $\{1, -\frac{1}{r}\}$, (W, B'_W) is sub-lc.

Let

$$\Delta' := \Delta + \pi_W^*(B'_W - B_W) \quad \text{and} \quad L'_W := L_W + B'_W - B_W = K_W + B'_W + M_W.$$

Then Δ' is a $\mathbb{Q}$-divisor and $K_{S\times W} + \Delta' \sim_{\mathbb{Q}} \pi_W^* L'_W$. Since $\Delta' = \Delta$ over $W \setminus \operatorname{Supp}(B'_W - B_W)$, π_W with the boundary Δ' is an lc-trivial fibration, which we denote by $\pi'_W : (S \times W, \Delta') \to W$, and we take L'_W as L in Definition 2.5. Let $f : W' \to W$ be a projective birational morphism from a normal variety. Since the pullbacks of $K_{S\times W} + \Delta'$ and $K_{S\times W} + \Delta$ to any higher model differ by the pullback of $f^*(B'_W - B_W)$, for any prime divisor P on W' the threshold t_P of Definition 2.5 for Δ' is the threshold for Δ minus $\operatorname{mult}_P f^*(B'_W - B_W)$. Let $\mathbf{B}$ and $\mathbf{B}'$ be the discriminant $\boldsymbol{b}$-divisors of π_W and π'_W, and $\mathbf{M}'$ the moduli $\boldsymbol{b}$-divisor of π'_W. Then $\mathbf{B}'_{W'} = \mathbf{B}_{W'} + f^*(B'_W - B_W)$, and since $\mathbf{M}$ descends to W,

$$\mathbf{M}'_{W'} = f^*L'_W - K_{W'} - \mathbf{B}'_{W'} = f^*L_W - K_{W'} - \mathbf{B}_{W'} = f^*M_W.$$

Thus the discriminant and the moduli $\boldsymbol{b}$-divisor of π'_W are B'_W and $\mathbf{M}$, so the moduli part of π'_W is M_W. Moreover, for any projective birational morphism $f : W' \to W$ from a normal variety,

$$K_{W'} + \mathbf{B}'_{W'} = f^*(K_W + B'_W). \tag{8.3}$$

We show that $(S \times W, \Delta')$ is sub-lc. Let v be a prime divisor over $S \times W$. If the center of v on $S \times W$ dominates W, then $a(v, S \times W, \Delta') \geq 0$ since $(S \times W, \Delta')$ is sub-lc over the generic point of W.

Suppose that the center of v on $S \times W$ does not dominate W. Let $X_1 \to S \times W$ be a projective birational morphism from a normal variety X_1 such that v is a prime divisor on X_1. The induced morphism $X_1 \to W$ is projective and surjective, and the generic fiber of $X_1 \to W$ is geometrically integral since it is birational to the generic fiber of the second projection π_W and S is a variety over the algebraically closed field $\mathbb{C}$. By [AK00, Theorem 2.1 and Proposition 4.4], there exist

projective modifications $X_2 \to X_1$ and $W_2 \to W$ and an equidimensional toroidal morphism $\pi_2 : X_2 \to W_2$ compatible with $X_1 \to W$. Since X_1 is normal, $X_2 \to X_1$ is an isomorphism over the generic point of v, so v is a prime divisor on X_2. Since π_2 is equidimensional,

$$\dim \pi_2(v) \geq \dim v - (\dim X_2 - \dim W_2) = \dim W_2 - 1,$$

and since v does not dominate W_2, $P := \pi_2(v)$ is a prime divisor on W_2 and $\operatorname{mult}_v \pi_2^* P$ is a positive integer. Let $K_{X_2} + \Delta_2$ be the pullback of $K_{S\times W} + \Delta'$ to X_2, and let t_P be the supremum of the real numbers t such that $(X_2, \Delta_2 + t\pi_2^* P)$ is sub-lc over the generic point of P. Since X_2 is birational over W_2 to the main component of $(S \times W) \times_W W_2$, $1 - t_P$ is the coefficient of P in the trace $\mathbf{B}'_{W_2}$ of the discriminant $\boldsymbol{b}$-divisor of π'_W. By (8.3) for $W_2 \to W$ and since (W, B'_W) is sub-lc,

$$1 - t_P = \operatorname{mult}_P \mathbf{B}'_{W_2} = 1 - a(P, W, B'_W) \leq 1,$$

so $t_P \geq 0$. For any real number $t < t_P$, we have

$$a(v, X_2, \Delta_2) - t \operatorname{mult}_v \pi_2^* P = a(v, X_2, \Delta_2 + t\pi_2^* P) \geq 0,$$

so

$$a(v, S \times W, \Delta') = a(v, X_2, \Delta_2) \geq t_P \operatorname{mult}_v \pi_2^* P \geq 0.$$

Thus $(S \times W, \Delta')$ is sub-lc. Moreover, by (8.2),

$$K_{S\times W} + \Delta' \sim_{\mathbb{Q}} \pi_W^*(K_W + B'_W + M_W) \sim_{\mathbb{Q}} 0.$$

Step 3. Let

$$Y := \operatorname{Spec}_W \bigoplus_{i\geq 0} \mathcal{O}_W(iA) \quad \text{and} \quad Z_0 := \operatorname{Spec} \bigoplus_{i\geq 0} H^0(W, iA),$$

so that Y is the total space of $\mathcal{O}_W(-A)$, with projection $q : Y \to W$ and zero section $E \cong W$. By [Kol13, 3.8], Z_0 is normal and the natural projective birational morphism $\mu : Y \to Z_0$ contracts E to the vertex o of Z_0 and is an isomorphism outside E. The variety Y is smooth, $\mathcal{O}_Y(E) \cong q^*\mathcal{O}_W(-A)$, and the relative cotangent sheaf of q is $q^*\mathcal{O}_W(A)$, so we may take canonical divisors with

$$K_Y + E = q^* K_W. \tag{8.4}$$

Let $X_0 := S \times Y$, let $\pi_Y : X_0 \to Y$ be the second projection, let $\pi_0 := \mu \circ \pi_Y : X_0 \to Z_0$, and let

$$B_0 := (\mathrm{id}_S \times q)^* \Delta' + S \times E.$$

Since $X_0 \to S \times W$ is the total space of the pullback of $\mathcal{O}_W(-A)$ with zero section $S \times E$, and $K_{X_0} + S \times E = (\mathrm{id}_S \times q)^* K_{S\times W}$ as in (8.4), (X_0, B_0) is sub-lc by **Step 2**. By (8.4) and **Step 2**,

$$K_{X_0} + B_0 = (\mathrm{id}_S \times q)^*(K_{S\times W} + \Delta') \sim_{\mathbb{Q}} 0.$$

Since π_Y and μ are contractions, π_0 is a contraction, and $\dim X_0 - \dim Z_0 = 2$. For any open subset $W_1 \subset W$ on which $\mathcal{O}_W(-A)$ is trivial,

$$q^{-1}(W_1) \backslash E \cong W_1 \times (\mathbb{A}^1 \backslash \{0\}),$$

over which $\pi_Y : (X_0, B_0) \to Y$ is the product of the restriction of π'_W over W_1 with $\mathbb{A}^1\backslash\{0\}$. Since π'_W is an lc-trivial fibration and μ is an isomorphism outside E, the product structure implies $\operatorname{rank} \pi_{0*}\mathcal{O}_{X_0}(\lceil \mathbf{A}^*(X_0, B_0) \rceil) = 1$.

Let $B_Y := E + q^* B'_W$. By (8.2) and (8.4),

$$K_Y + B_Y + q^* M_W = q^*(K_W + B'_W + M_W) = \frac{1}{r} \operatorname{div}_Y(\varphi). \tag{8.5}$$

Let $K_{Z_0} := \mu_* K_Y$, $B_{Z_0} := \mu_* q^* B'_W$, and $M_{Z_0} := \mu_* q^* M_W$. Applying μ_* to (8.5), we obtain

$$L_0 := K_{Z_0} + B_{Z_0} + M_{Z_0} = \frac{1}{r} \operatorname{div}_{Z_0}(\varphi),$$

so L_0 is $\mathbb{Q}$-Cartier, $K_{X_0}+B_0 \sim_{\mathbb{Q}} 0 \sim_{\mathbb{Q}} \pi_0^* L_0$, and

$$K_Y + B_Y + q^* M_W = \mu^*(K_{Z_0} + B_{Z_0} + M_{Z_0}). \tag{8.6}$$

Thus $\pi_0 : (X_0, B_0) \to Z_0$ is an lc-trivial fibration with B_0 a $\mathbb{Q}$-divisor, and we take L_0 as L in Definition 2.5.

We show that B_Y is the trace on Y of the discriminant $\boldsymbol{b}$-divisor of π_0. Since π_0 factors through π_Y, the trace on Y of the discriminant $\boldsymbol{b}$-divisor of π_0 is the discriminant of $\pi_Y : (X_0, B_0) \to Y$. Let P be a prime divisor on Y and t_P the threshold of Definition 2.5 for π_Y.

- If $P = E$, then $\pi_Y^* E = S \times E$ has coefficient 1 in B_0 and (X_0, B_0) is sub-lc, so $t_E = 0$ and $\operatorname{mult}_E B_Y = 1$.
- If P does not dominate W, then $P = q^* P_W$ for the prime divisor $P_W := q(P)$ on W. The generic point of P is not in E, so by the product structure over $q^{-1}(W_1)\backslash E$ for such W_1 meeting P_W, $1 - t_P$ is the coefficient of P_W in the discriminant B'_W of π'_W, which is $\operatorname{mult}_P q^* B'_W$.
- If P dominates W and $P \neq E$, then over the generic point of W, π_Y is the product of the generic fiber of π'_W with the fiber of q, and P corresponds to a closed point of that fiber other than the zero point. Since the generic fiber of π'_W is sub-lc, $t_P = 1$ and $\operatorname{mult}_P B_Y = 0$.

Thus the trace on Y is B_Y. Since μ is an isomorphism outside E and E is μ-exceptional, the discriminant of π_0 is $\mu_* B_Y = B_{Z_0}$, and the moduli part of π_0 is

$$L_0 - K_{Z_0} - B_{Z_0} = M_{Z_0}.$$

Step 4. In this step we construct $\pi : (X, B) \to Z$ and z, and we prove (1)–(3).

Let s be the coordinate of $\mathbb{A}^1$, and let

$$Z := Z_0 \times \mathbb{A}^1, \quad z := (o, 0), \quad X := X_0 \times \mathbb{A}^1, \quad \text{and} \quad \pi := \pi_0 \times \mathrm{id}_{\mathbb{A}^1} : X \to Z.$$

Let B be the pullback of B_0 to X. By **Step 3**, $\pi : (X, B) \to Z$ is an lc-trivial fibration with B a $\mathbb{Q}$-divisor, z is a closed point of Z, and $\dim X - \dim Z = 2$. We take $L := \mathrm{pr}_{Z_0}^* L_0$ and $K_Z := \mathrm{pr}_{Z_0}^* K_{Z_0}$. For a prime divisor P on Z, either $P = \mathrm{pr}_{Z_0}^* P_0$ for a prime divisor P_0 on Z_0, and then $t_P = t_{P_0}$, or P dominates Z_0, and then $t_P = 1$ since over the generic point of Z_0, (X, B) is the product of the generic fiber of π_0 with $\mathbb{A}^1$ and $\pi^* P$ is reduced. Thus $B_Z = \mathrm{pr}_{Z_0}^* B_{Z_0}$ and $M_Z = \mathrm{pr}_{Z_0}^* M_{Z_0}$. Let $\mathrm{pr} : Y \times \mathbb{A}^1 \to Y$ be the first projection and $K_{Y\times\mathbb{A}^1} := \mathrm{pr}^* K_Y$. By (8.6),

$$(\mu \times \mathrm{id}_{\mathbb{A}^1})^*(K_Z + B_Z + M_Z) = K_{Y\times\mathbb{A}^1} + \mathrm{pr}^* B_Y + \mathrm{pr}^* q^* M_W. \tag{8.7}$$

By **Step 3**, (X_0, B_0) is sub-lc, so $(X, B + \pi^*(s = 0))$ is sub-lc. Let v be a prime divisor over X with $\pi(\operatorname{center}_X v) = \{z\}$. Since $\operatorname{center}_X v \subset \pi^{-1}(z) \subset \operatorname{Supp} \pi^*(s = 0)$, $\operatorname{mult}_v \pi^*(s = 0) \geq 1$, so

$$a(v, X, B) = a(v, X, B + \pi^*(s = 0)) + \operatorname{mult}_v \pi^*(s = 0) \geq 1.$$

Thus $\operatorname{mld}(X/Z \ni z, B) \geq 1$. The generic fiber of π is the base change of S to the generic point of Z, which is of Fano type (Definition 8.2) since S is klt and $-K_S$ is ample. Thus (1)–(3) hold.

Step 5. In this step we prove (4) and the last sentence of Corollary 1.4.

Let $V \subset Z$ be an open neighborhood of z and $D \geq 0$ an $\mathbb{R}$-divisor on V with $D \sim_{\mathbb{R}} M_Z|_V$, and write

$$D - M_Z|_V = \sum_{i \in I} \nu_i \operatorname{div}(\rho_i)$$

with I a finite set, $\nu_i \in \mathbb{R}$, and ρ_i rational functions on V. Let $V' := (\mu \times \mathrm{id}_{\mathbb{A}^1})^{-1}(V)$, which is smooth. Since $K_Z + B_Z + M_Z$ is $\mathbb{Q}$-Cartier, $K_V + B_Z|_V + D$ is $\mathbb{R}$-Cartier, and we define the $\mathbb{R}$-divisor D' on V' by

$$(\mu \times \mathrm{id}_{\mathbb{A}^1})^*(K_V + B_Z|_V + D) = K_{V'} + \mathrm{pr}^* B_Y|_{V'} + D'.$$

By (8.7),

$$D' = \mathrm{pr}^*\, q^* M_W|_{V'} + \sum_{i\in I} \nu_i \,\mathrm{div}_{V'}(\rho_i) \sim_{\mathbb{R}} \mathrm{pr}^*\, q^* M_W|_{V'}.$$

Since $\mu \times \mathrm{id}_{\mathbb{A}^1}$ is an isomorphism outside $E \times \mathbb{A}^1$, there exists a real number α with

$$D' = \widetilde{D} + \alpha(E \times \mathbb{A}^1)|_{V'}, \quad \text{where } \widetilde{D} := (\mu \times \mathrm{id}_{\mathbb{A}^1})_*^{-1} D \geq 0.$$

Let $B_{V'} := \mathrm{pr}^*\, B_Y|_{V'} + D'$. Then $a(v, V, B_Z|_V + D) = a(v, V', B_{V'})$ for any prime divisor v over V', and if $\mathrm{center}_{V'}\, v \subset E \times \{0\}$, then $\mathrm{center}_V\, v = \{z\}$.

Let $C := \{\tau \in \mathbb{A}^1 \mid (o, \tau) \in V\}$, an open neighborhood of 0, and for $\tau \in C$, $E \times \{\tau\} \subset V'$ is the fiber of $\mu \times \mathrm{id}_{\mathbb{A}^1}$ over (o, τ) and is identified with W by q. For general $\tau \in C$, $E \times \{\tau\} \not\subset \mathrm{Supp}\, \widetilde{D}$, and restricting the $\mathbb{R}$-linear equivalence class of D' to $E \times \{\tau\}$, since $(E \times \mathbb{A}^1)|_{E\times\{\tau\}} \sim -A$, we obtain

$$\widetilde{D}|_{E\times\{\tau\}} \sim_{\mathbb{R}} M_W + \alpha A. \tag{8.8}$$

If $\alpha < 0$, then $M_W \sim_{\mathbb{R}} \widetilde{D}|_{E\times\{\tau\}} + (-\alpha)A$ with $\widetilde{D}|_{E\times\{\tau\}} \geq 0$ and $(-\alpha)A$ ample, so M_W is big, contradicting **Step 1**. Thus $\alpha \geq 0$.

Suppose that $\alpha > 0$. Let w_0 be a local equation of $E \times \mathbb{A}^1$ at the generic point of $E \times \{0\}$. Then w_0, s form a regular system of parameters there. For any prime divisor P_W on W, $q^* P_W \not\supset E$, so the only component of $\mathrm{pr}^*\, B_Y$ containing $E \times \{0\}$ is $E \times \mathbb{A}^1$, of coefficient 1. For any positive integer k, let v_k be the monomial valuation in w_0, s with weights $(k, 1)$. Then v_k is divisorial and $\mathrm{center}_{V'}\, v_k = E \times \{0\}$. Since $\mathrm{mult}_{E\times\mathbb{A}^1} B_{V'} = 1 + \alpha$ and $\widetilde{D} \geq 0$,

$$a(v_k, V', B_{V'}) \leq k + 1 - (1 + \alpha)k = 1 - k\alpha.$$

Suppose that $\alpha = 0$. By (8.8) and **Step 1**, $p \in \mathrm{Supp}(\widetilde{D}|_{E\times\{\tau\}})$ for general $\tau \in C$, where we regard p as a point of $E \times \{\tau\}$. Since $\mathrm{Supp}\, \widetilde{D}$ is closed in V', it contains the curve $\{p\} \times C$. Let $\beta > 0$ be the sum of the coefficients in $\widetilde{D}$ of the components of $\widetilde{D}$ containing $\{p\} \times C$. At $(p, 0)$, let w_0 be a local equation of $E \times \mathbb{A}^1$ and w_j a local equation of $\mathrm{pr}^*\, q^* R_j$ for $j = 1, \ldots, 5$. Since $R_1, \ldots, R_5$ intersect transversely at p, $w_0, \ldots, w_5, s$ form a regular system of parameters of V' at $(p, 0)$, and $\{p\} \times C = (w_0 = \cdots = w_5 = 0)$ near $(p, 0)$. Since $p \notin \mathrm{Supp}\, F$, near $(p, 0)$ we have

$$B_{V'} = \sum_{j=0}^{5} (w_j = 0) + \widetilde{D}.$$

For any positive integer k, let v_k be the monomial valuation in $w_0, \ldots, w_5, s$ with weights $(k, \ldots, k, 1)$. Then v_k is divisorial and $\mathrm{center}_{V'}\, v_k = \{(p, 0)\} \subset E \times \{0\}$. A local equation of any component of $\widetilde{D}$ containing $\{p\} \times C$ is in the ideal $(w_0, \ldots, w_5)$, so has value $\geq k$ under v_k. Thus

$$a(v_k, V', B_{V'}) \leq 6k + 1 - 6k - k\beta = 1 - k\beta.$$

For $\alpha > 0$ and for $\alpha = 0$, letting $k \to +\infty$, we obtain $\mathrm{mld}(V \ni z, B_Z|_V + D) = -\infty$. Thus (4) holds. If Conjecture 1.2 held for $d = 2$ and $\epsilon = 1$, then by (1)–(3) there would be a real number $\delta > 0$ and an $\mathbb{R}$-divisor $D \geq 0$ on Z with $D \sim_{\mathbb{R}} M_Z$ and $\mathrm{mld}(Z \ni z, B_Z + D) \geq \delta > 0$, contradicting (4) with $V = Z$. □

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